Présentation IUT Agile Mai 2014
#iutagile http://iutagile.com

Continuous Integration

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This document:
http://arnaud.nauwynck.free.fr/IUT-Agile-2014-ContinuousInt.pdf

Summary

- What is Continuous Integration ?
- Presentation of Tools Ecosystem
- Live Demo with...
  - Test Driven (TDD) perspective
  - Most important Tools
  - Focusing on Java
  - In 1h30 !!! Fasten Seat Belts ….

Who I Am

- IT Passionate
- IT Architect since ~7 years
- Working at SGCIB – Lyxor, Paris - La Défense
- Doing Archi, Dev, Tools, DevOps... on Java/C#
- Prof at IUT ~ 15 days/year

Continuous + Integration

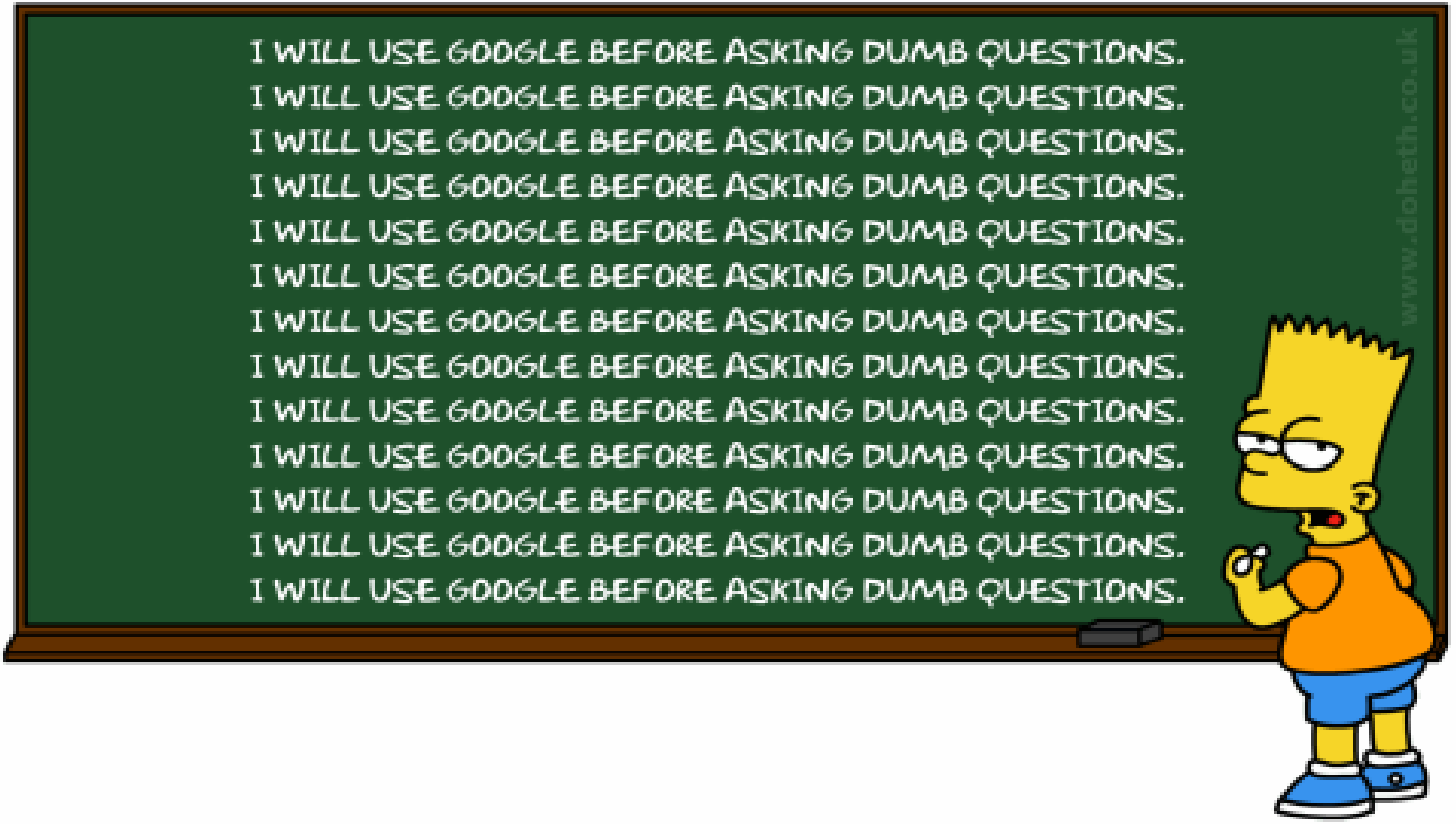
In French : “Intégration Continue” …

2 Math terms … with precise semantics

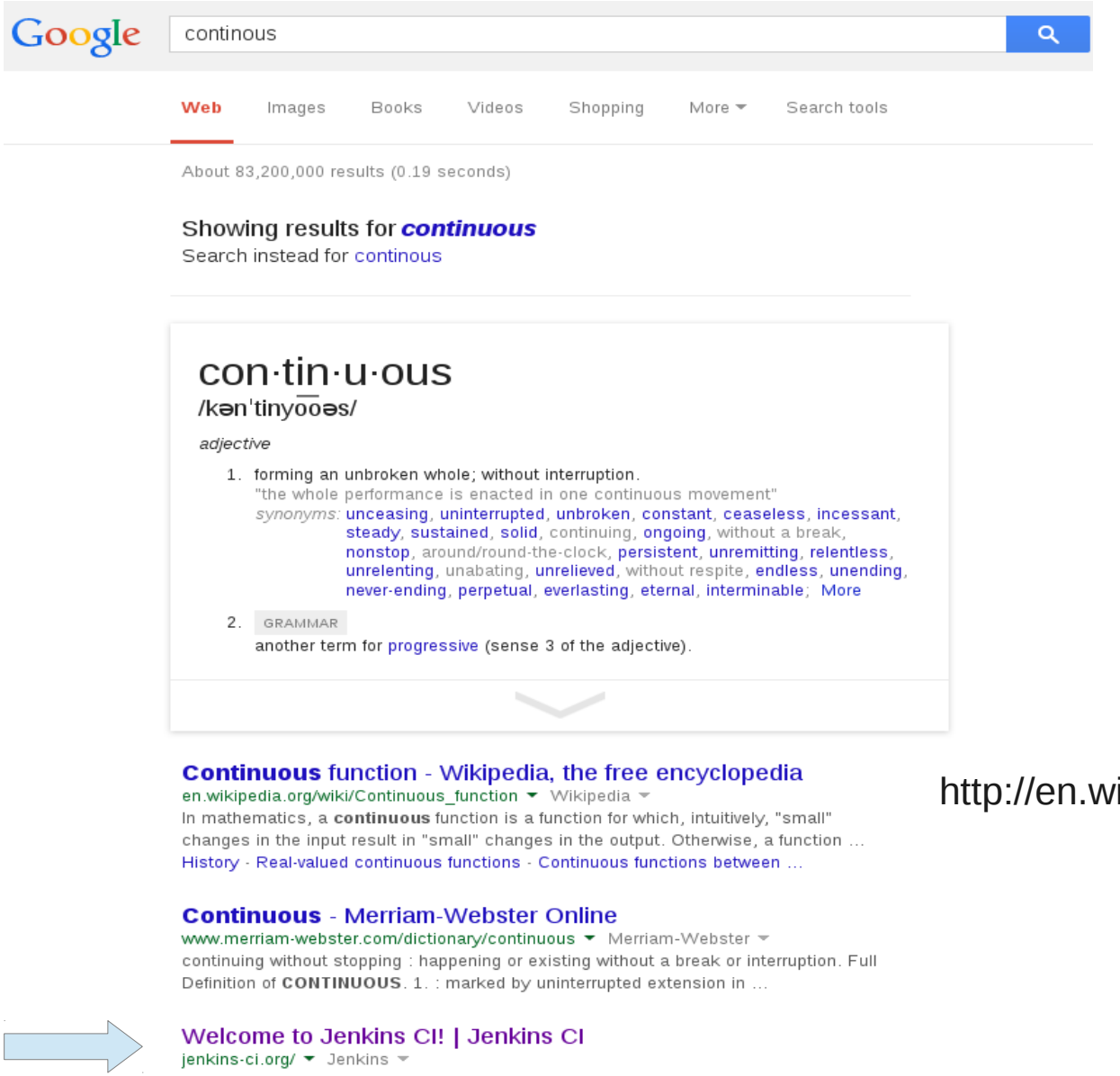
2 “common” terms

In Computer science ??
= Automated Build,Test & Deployment … Tools

What is “Continuous Integration” ?



“Continuous” Google rank #3 = Jenkins



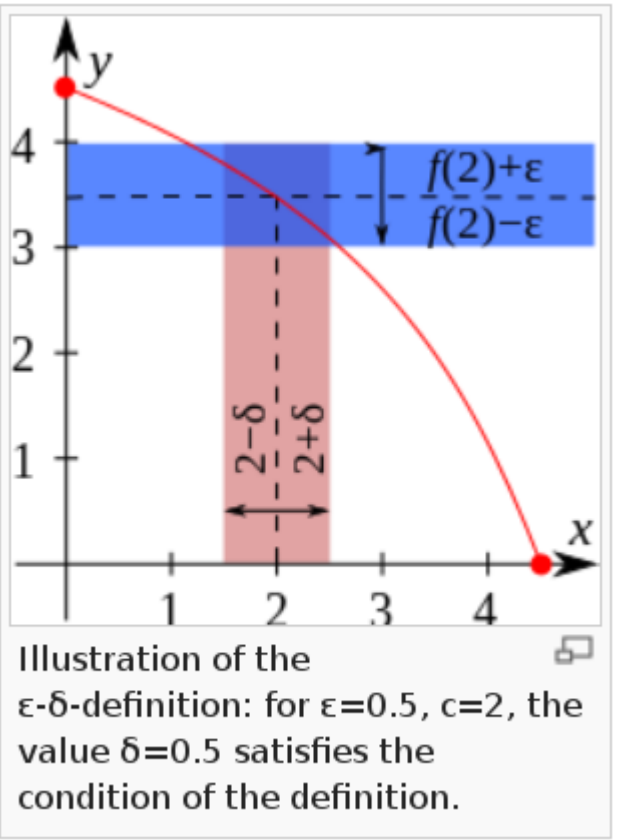


Illustration of the ε-δ-definition: for ε=0.5, c=2, the value δ=0.5 satisfies the condition of the definition.

http://en.wikipedia.org/wiki/Continuous_function

“Integration” Google rank #4 = Jenkins !

Google integration +Arnaud

Web Images Books Videos News More Search tools

About 113,000,000 results (0.28 seconds)

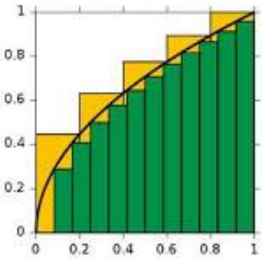
Integration - Wikipedia, the free encyclopedia
en.wikipedia.org/wiki/**Integration** Wikipedia
Integration may refer to: Contents. 1 Sociology and economy; 2 Mathematics; 3 Electronics engineering; 4 Genetics/Enzymology; 5 Other uses; 6 See also ...
Integral · Social integration · Racial integration · System integration

Integration testing - Wikipedia, the free encyclopedia
en.wikipedia.org/wiki/**Integration**_testing Wikipedia
Integration testing (sometimes called **integration** and testing, abbreviated I&T) is the phase in software testing in which individual software modules are ...

Integration | Define **Integration** at Dictionary.com
dictionary.reference.com/browse/**integration**
an act or instance of combining into an integral whole. 2. an act or instance of **integrating** a racial, religious, or ethnic group. 3. an act or instance of **integrating** an ...
Integrationist · Integration by parts · Contour integration · Constant of integration

Welcome to Jenkins CI! | Jenkins CI
jenkins-ci.org/ Jenkins

Integral



Integration is an important concept in mathematics and, together with its inverse, differentiation, is one of the two main operations in calculus. Wikipedia

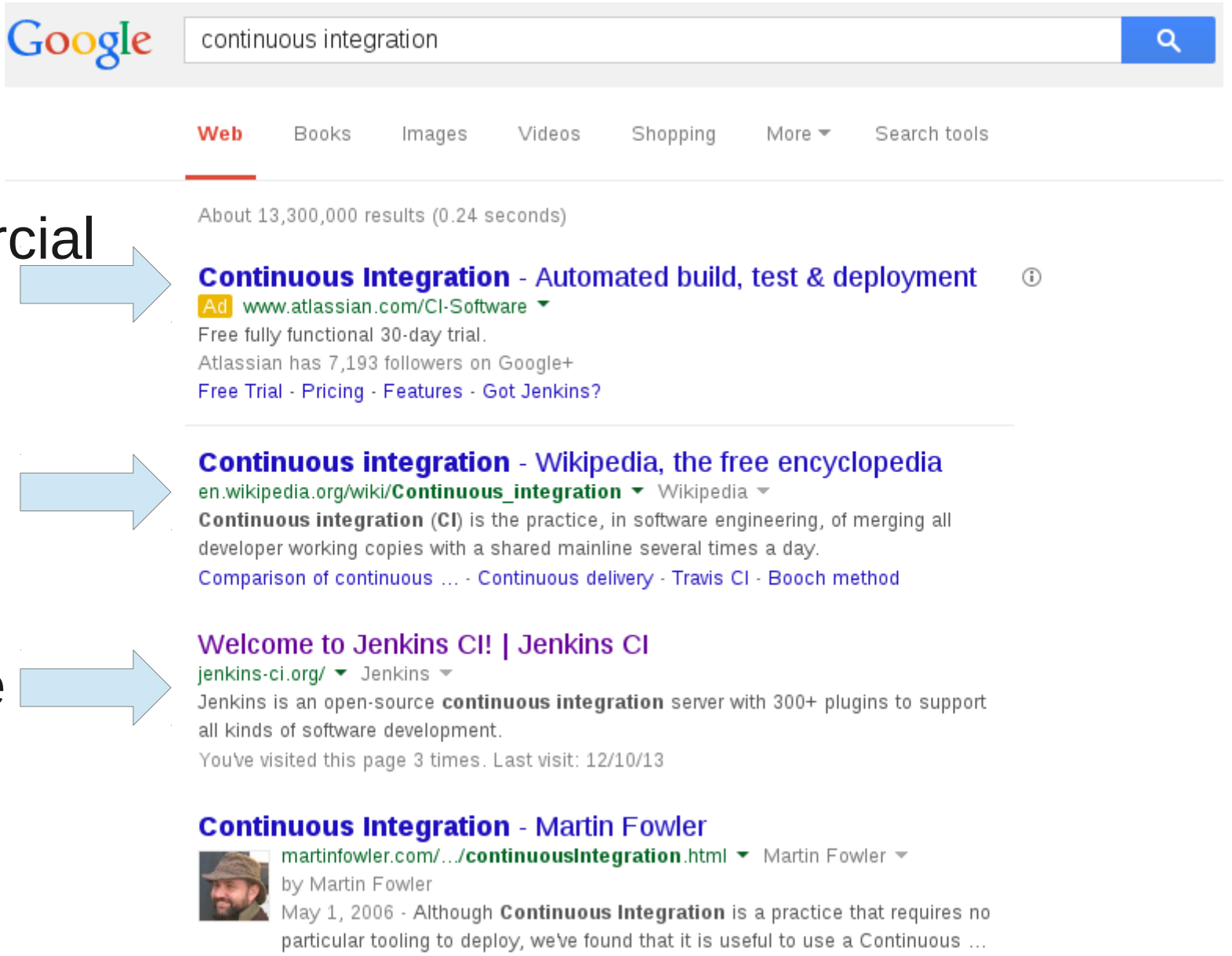
Related topics

The term integral may also refer to the related notion of the antiderivative, a function F whose **derivative** is the given function f. Wikipedia
Explore: Derivative

Calculus, historically known as infinitesimal calculus, is a mathematical discipline focused on limits, functions, **derivatives**, integrals, and infinite series. Wikipedia
Explore: Calculus, Derivative

The integrals encountered in a basic **calculus** course are deliberately chosen for simplicity; those found in **real** applications are not always so accommodating. Wikipedia
Explore: Calculus, Real number

“Continuous Integration ” Google



Commercial #1

Theory #2

Practise #3

Wikipedia



WIKIPEDIA
The Free Encyclopedia

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Continuous integration

From Wikipedia, the free encyclopedia

Main page
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Current events
Random article

Continuous integration (**CI**) is the practice, in software engineering, of merging all developer working copies with a shared mainline several times a day. It was first named and proposed as part of extreme programming (XP). Its main aim is to prevent integration problems,

CI was originally intended to be used in combination with automated unit tests written through the practices of test-driven development.

Merge Work - */Day - Mainline - Test



Code

Mainline



Automated
Build + Test

Test Test Test Test

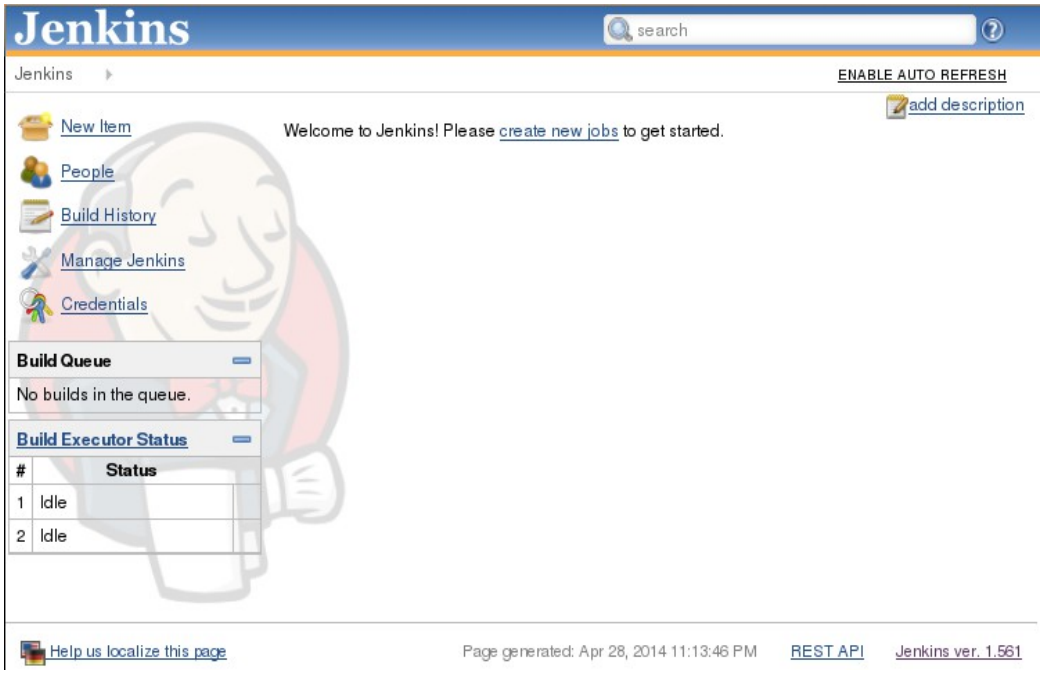
Meet Jenkins

Jenkins

http://jenkins-ci.org

http://localhost:8080/jenkins





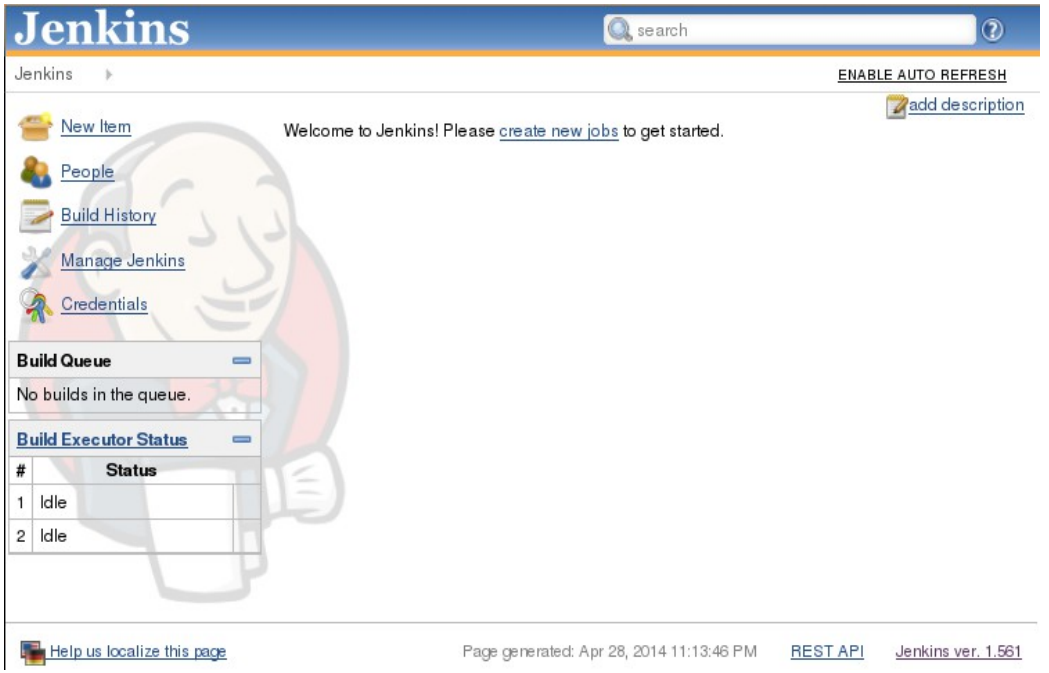
Meet Jenkins

Jenkins

http://jenkins-ci.org

http://localhost:8080/jenkins





What

http://jenkins-ci.org

"An extendable open source **continuous integration server**"

Get Project Source Code







Build Project Output

Archive Build results,
Test reports,
Emails ...

How

Jenkins

"An **extendable** open source continuous integration server"



Jobs & Plugins



Get Project
Source Code

Trigger
from SVN, GIT,
Or other events

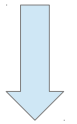






Build Project Output

Export Build results
Html, reports,
Emails ...





Launch "builds"
Processes
(maven, ant,
.cmd, graddle, ...)

Get Hands Dirty

Start as
- tomcat
- standalone jar

Tomcat/bin/statup.sh

java -jar jenkins.war

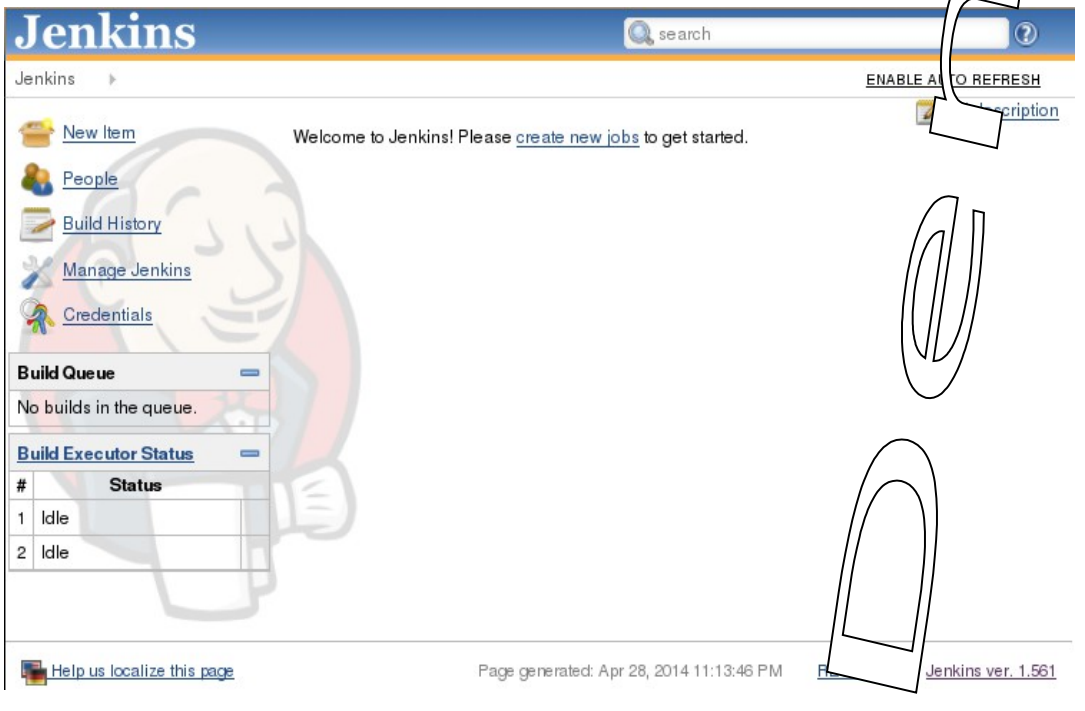
Check log OK

```
INFO: Started SelectChannelConnector@0.0.0.0:8080
Apr 28, 2014 11:12:36 PM winstone.Logger logInternal

Apr 28, 2014 11:12:43 PM hudson.WebAppMain$3
INFO: Jenkins is fully up and running
Effective SlaveRestarter on master: null
```

http://localhost:8080

Open Browser

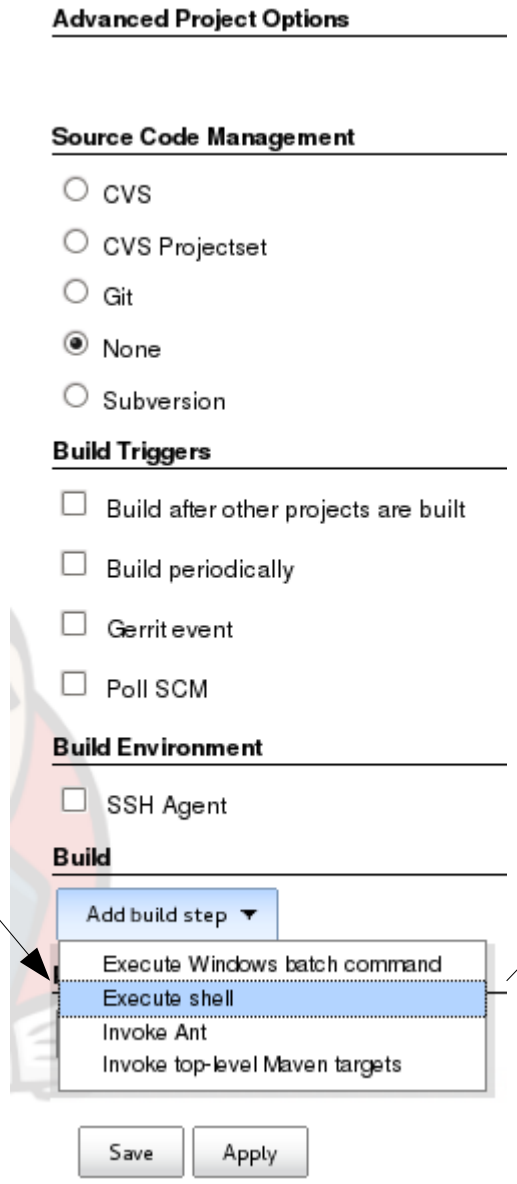


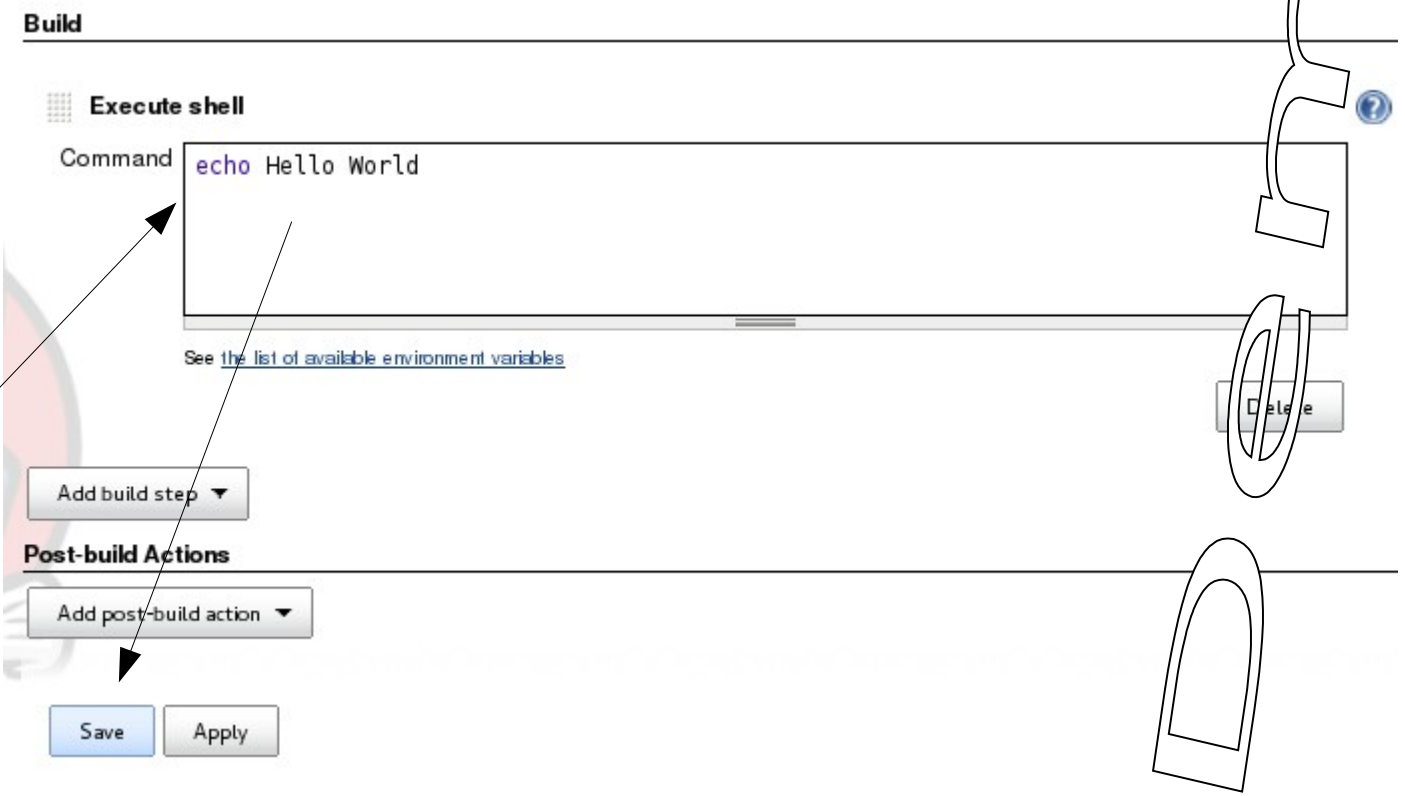
Jenkins “Hello World”
Get Hands Dirty



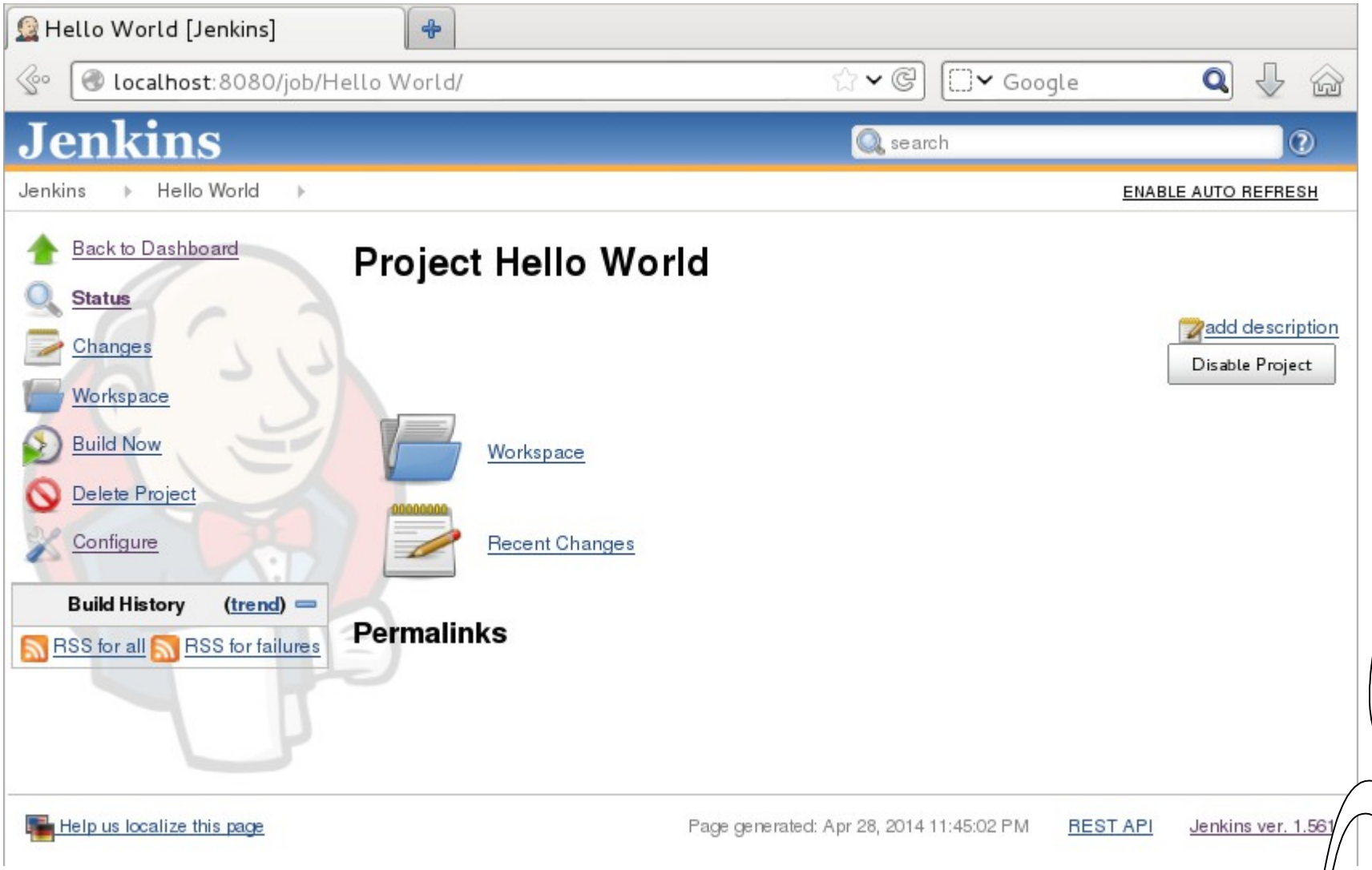
Jenkins

- Add Build Step “Execute Shell”
- Enter shell text
  echo Hello World
- click SAVE





Jenkins > Job


Hello World [Jenkins]
localhost:8080/job/Hello World/
Google
Jenkins
search
Jenkins
Hello World
ENABLE AUTO REFRESH
Back to Dashboard
Status
Changes
Workspace
Build Now
Delete Project
Configure
Project Hello World
add description
Disable Project
Workspace
Recent Changes
Build History
(trend)
RSS for all
RSS for failures
Permalinks
Help us localize this page
Page generated: Apr 28, 2014 11:45:02 PM
REST API
Jenkins ver. 1.561

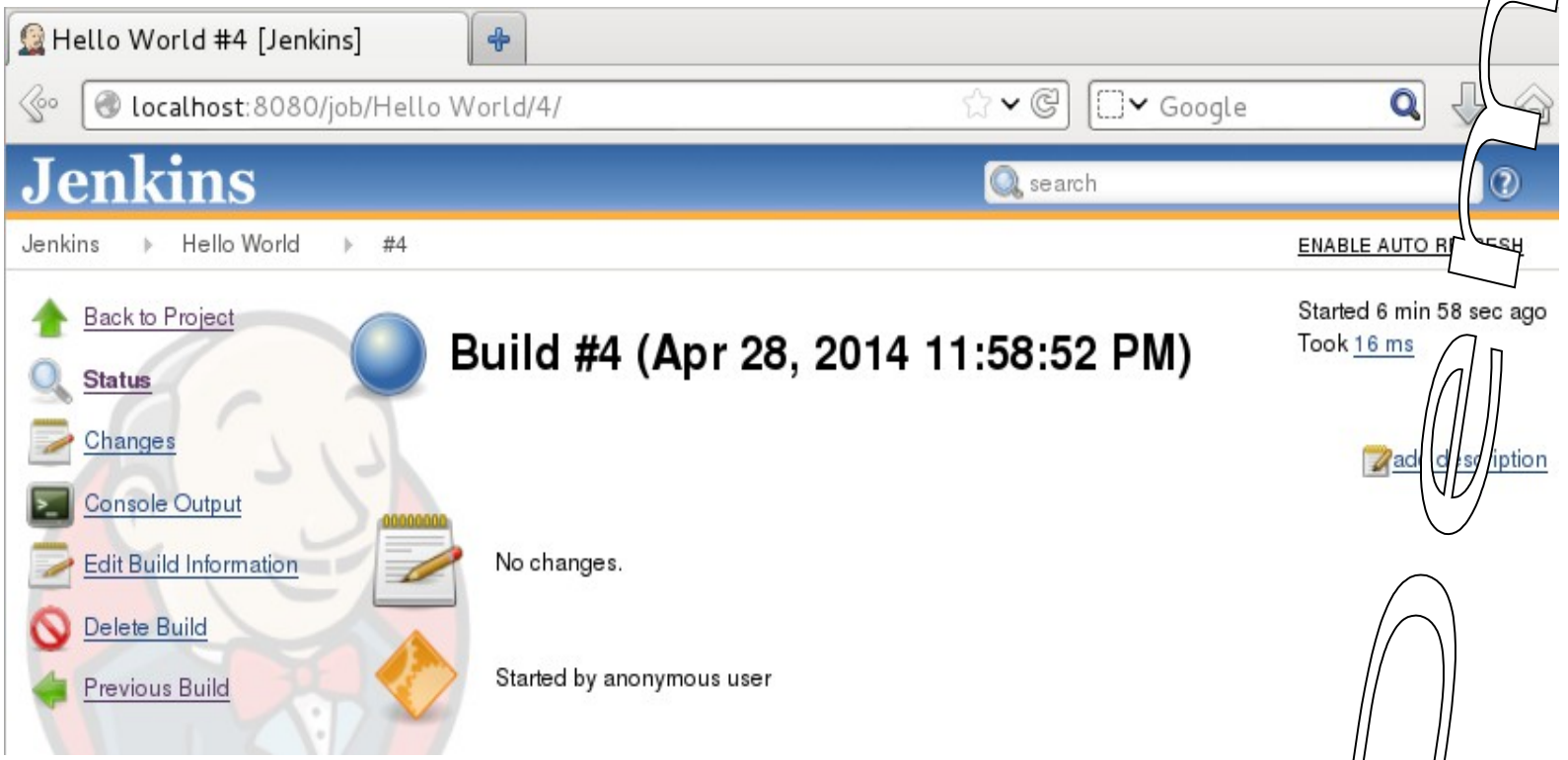
Build Now (3/3)
Build = instance of launching a Job



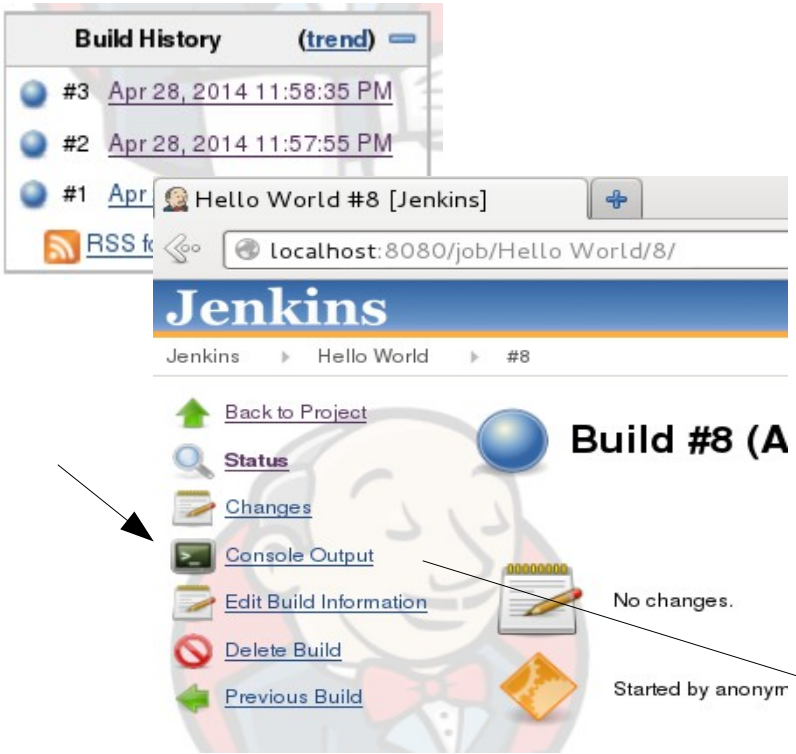
Demo

Jenkins > Job > Build





> Job > Build # > Console





Navigation

Dashboard > Job > Build > Result

URL = Navigation Toolbar = Navigation into jenkins object





Contextual Action / Menu

Jenkins

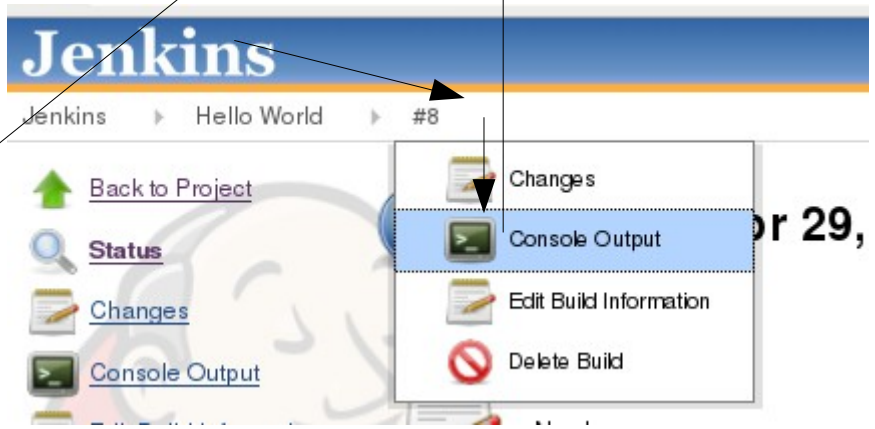
From Object / Action Menu





Directly From Context Menu





Is That All ? Is It Used ?

Jenkins

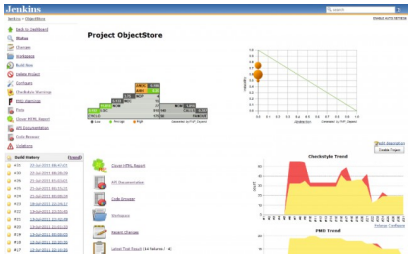
NOT just a CRON !!!



Plugins => Huge Value Added

… on reports

… Integration of MANY Tools



Build History => Reports, Stats, Trends, ...



De Factor Standard – in Industry & Open-Source

Ultra Active



How Much Extensible Configurable ?

Jenkins

Cf http://jenkins-ci.org



Customize Jenkins

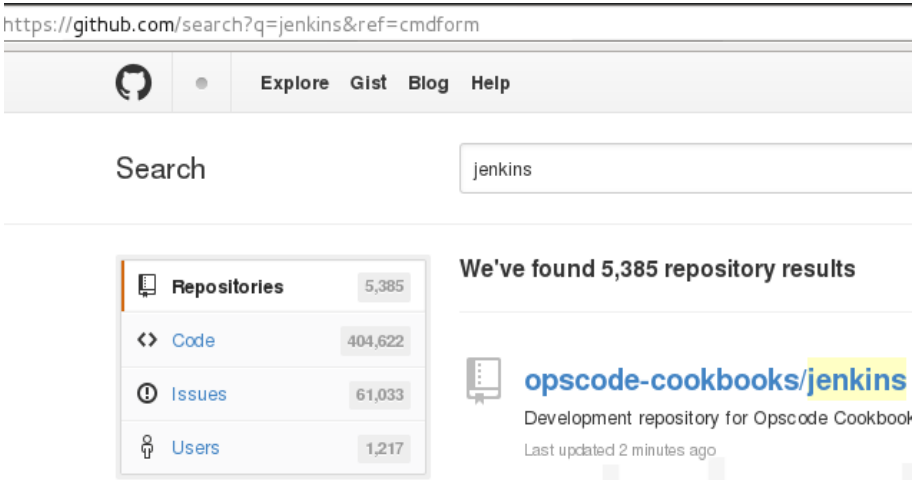
Choose from over **600** plugins to customize Jenkins exactly to your needs.

600 Plugins !



Configurable + extensible

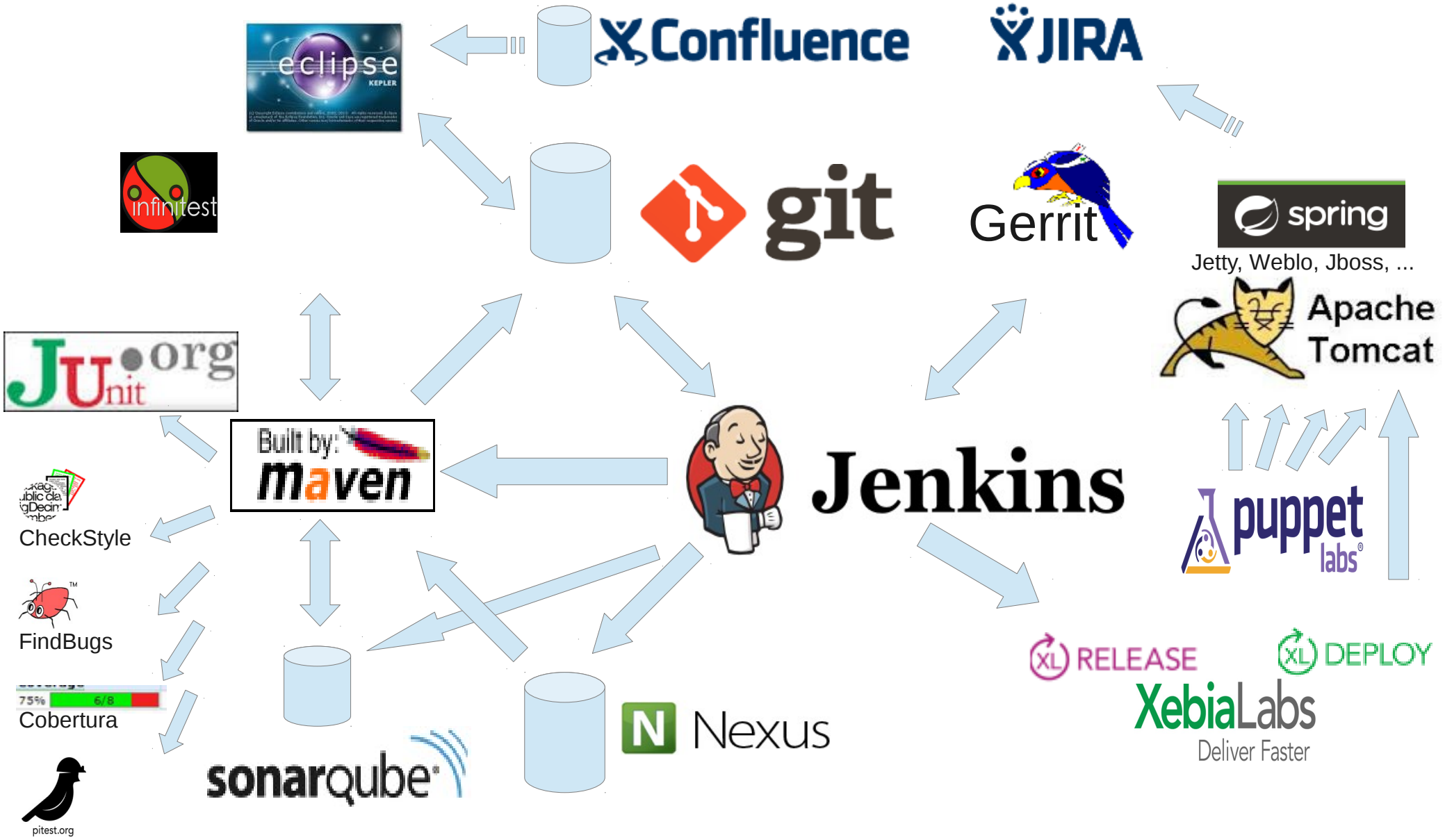
https://github.com/jenkinsci



Fork me on GitHub

5385 GitHub Repos

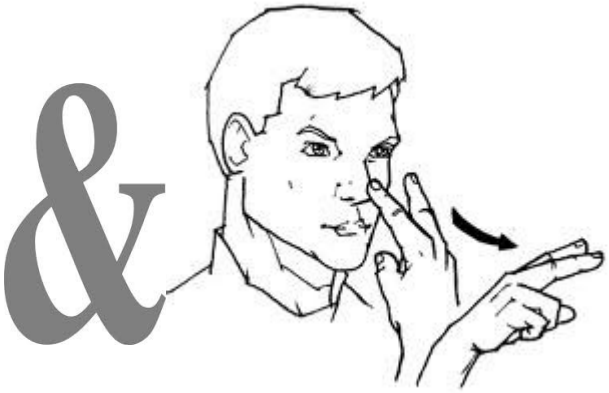
Tools Ecosystem / Plugins



Don't Panic...





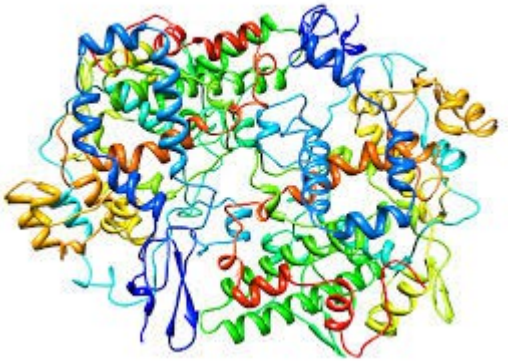


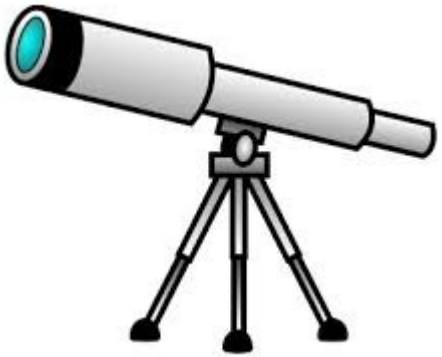
First Review It … Several Ways







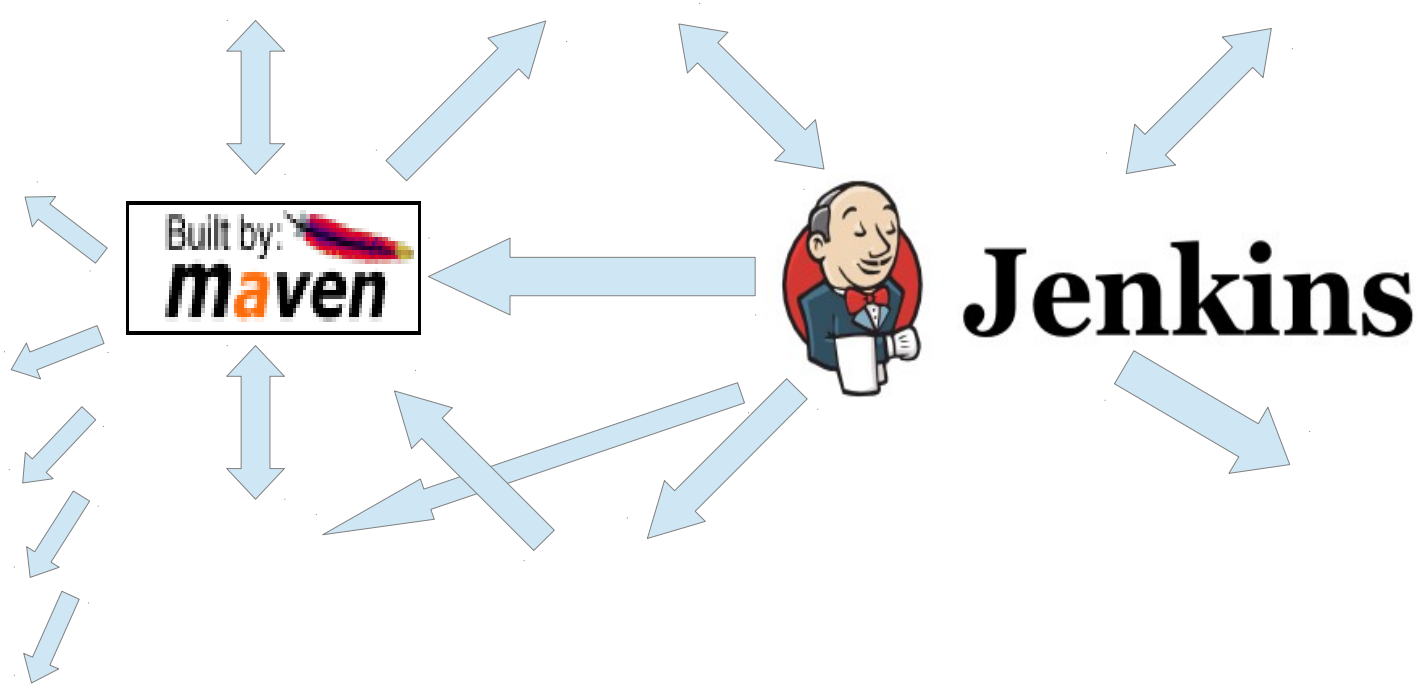






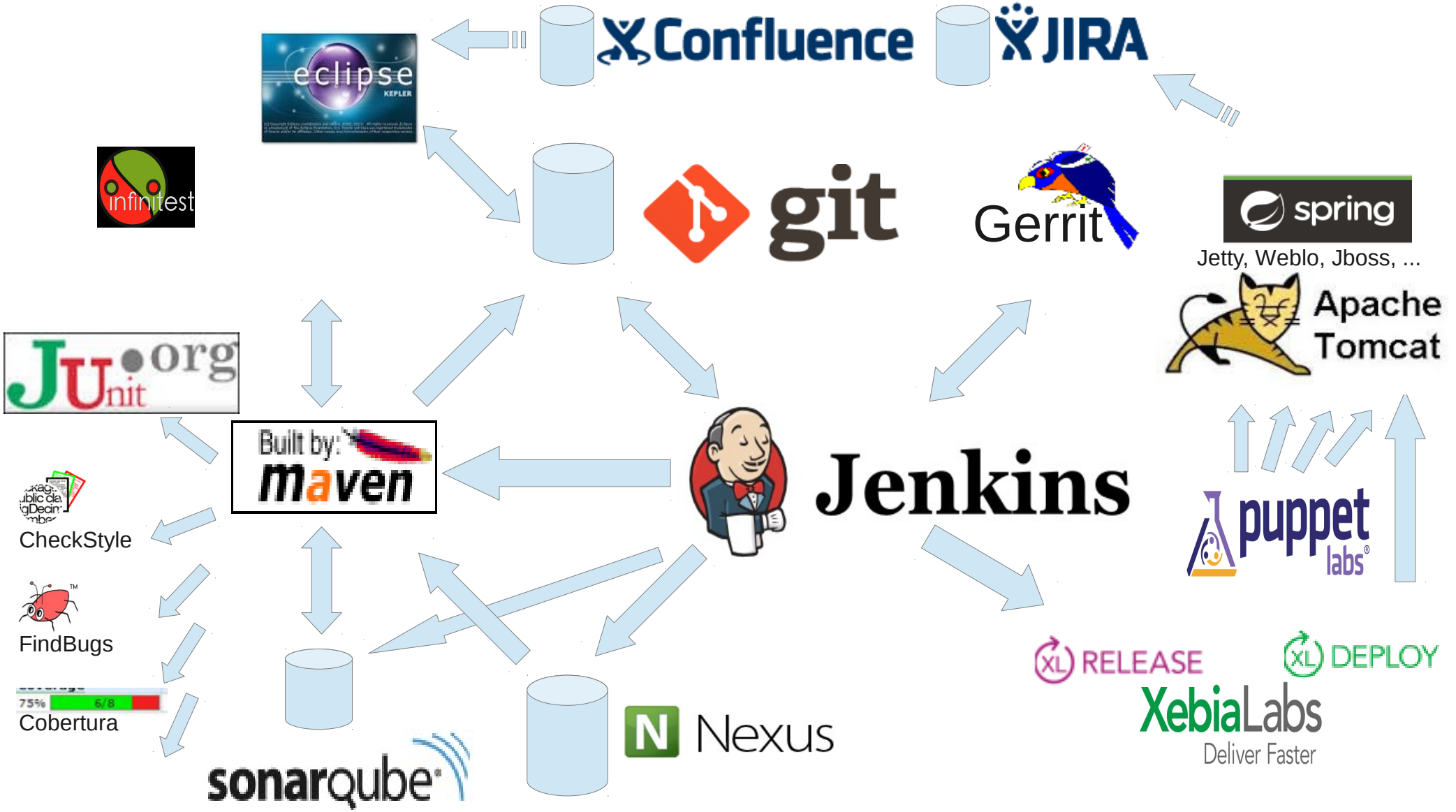


CORE Interconnected Tools



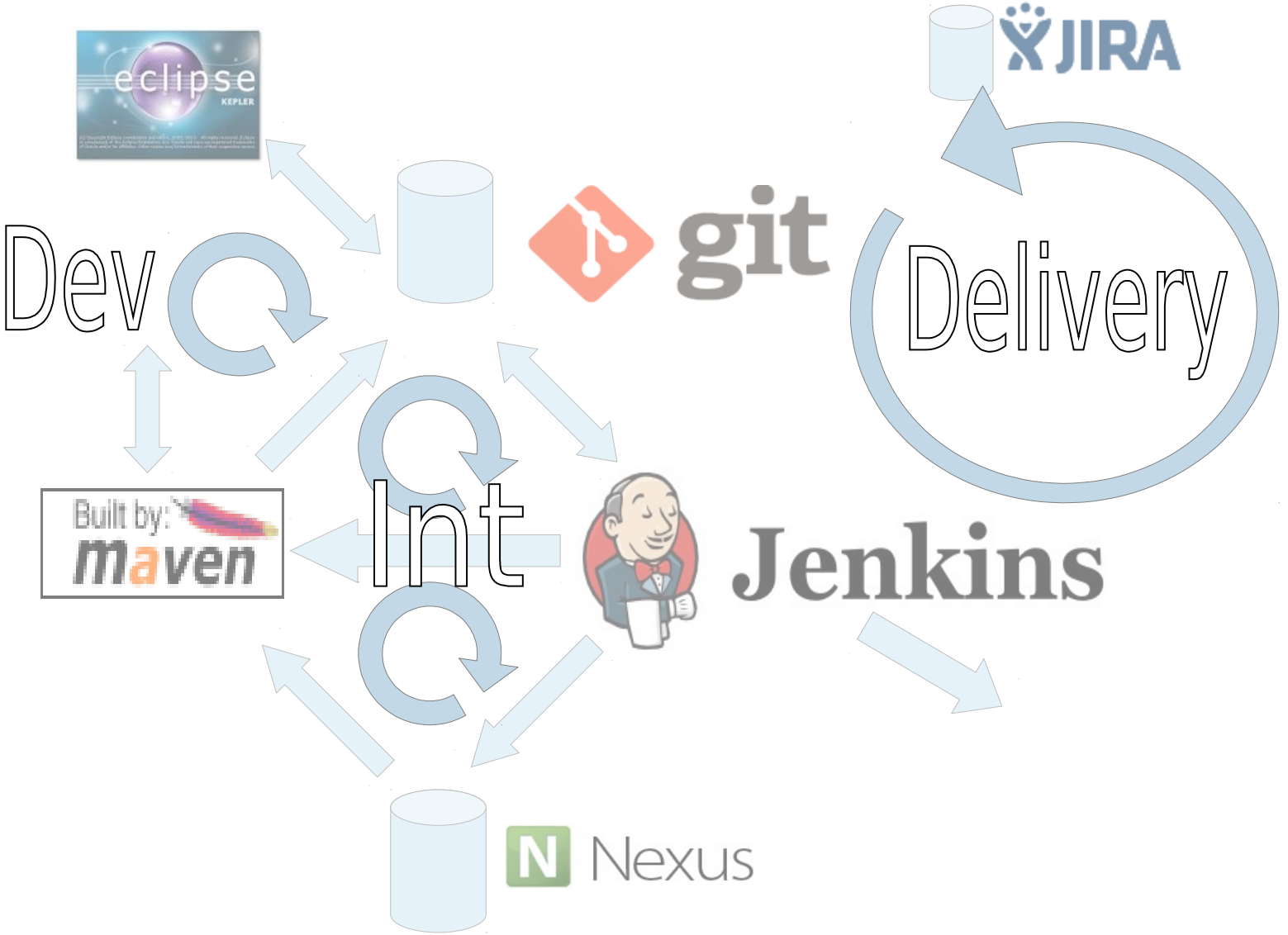
Quizz ...
Find Circular Loop ?



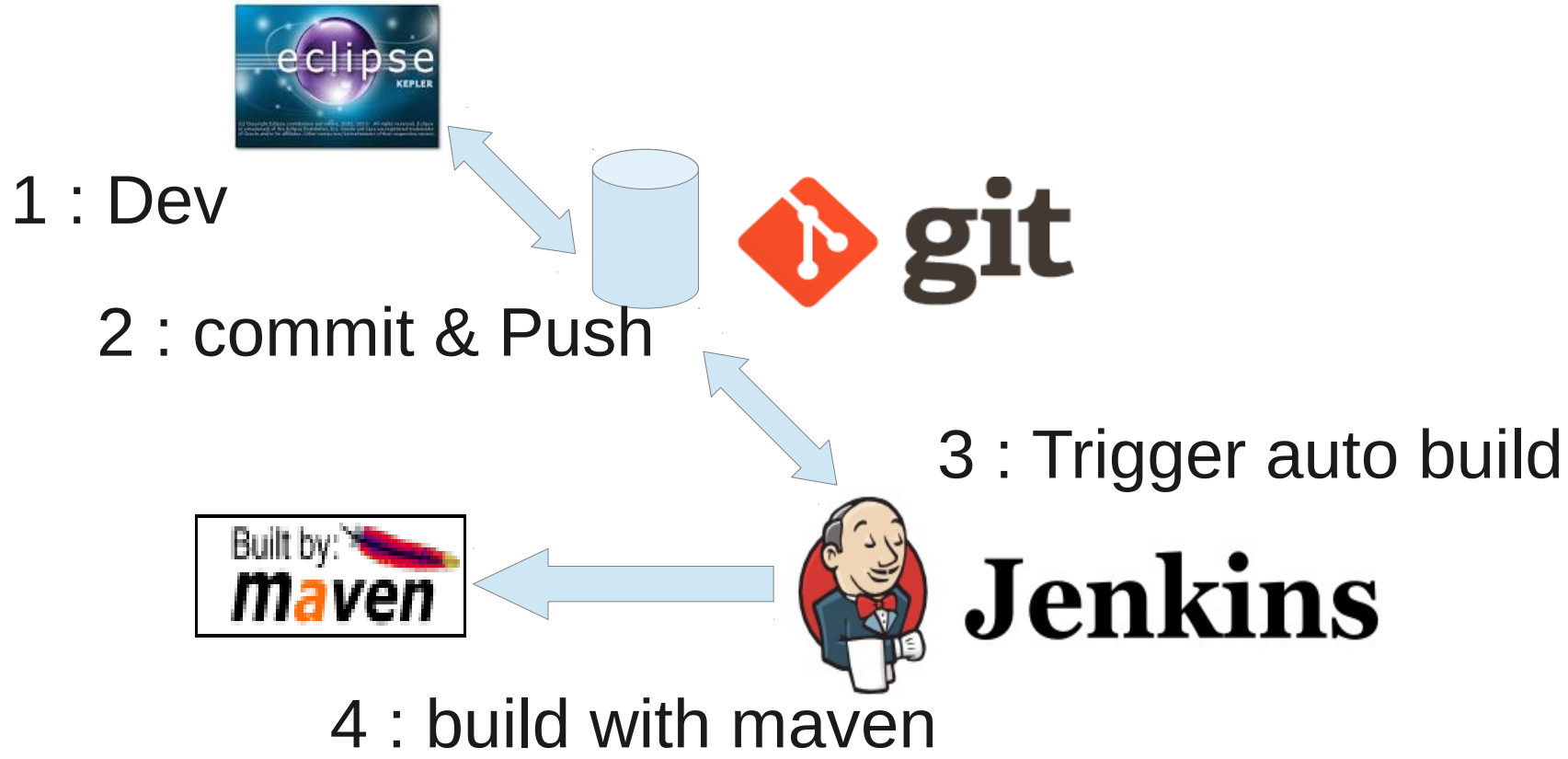


Core Workflow
Dev → Integration → Delivery





Day to Day Development Workflow



Divide in Groups ?





Confluence

JIRA



git

Gerrit

spring

Jetty, Weblo, Jboss, ...

Apache Tomcat

JUnit.org

Built by: maven

Jenkins

puppet labs

CheckStyle

FindBugs

Cobertura

sonarqube

Nexus

RELEASE

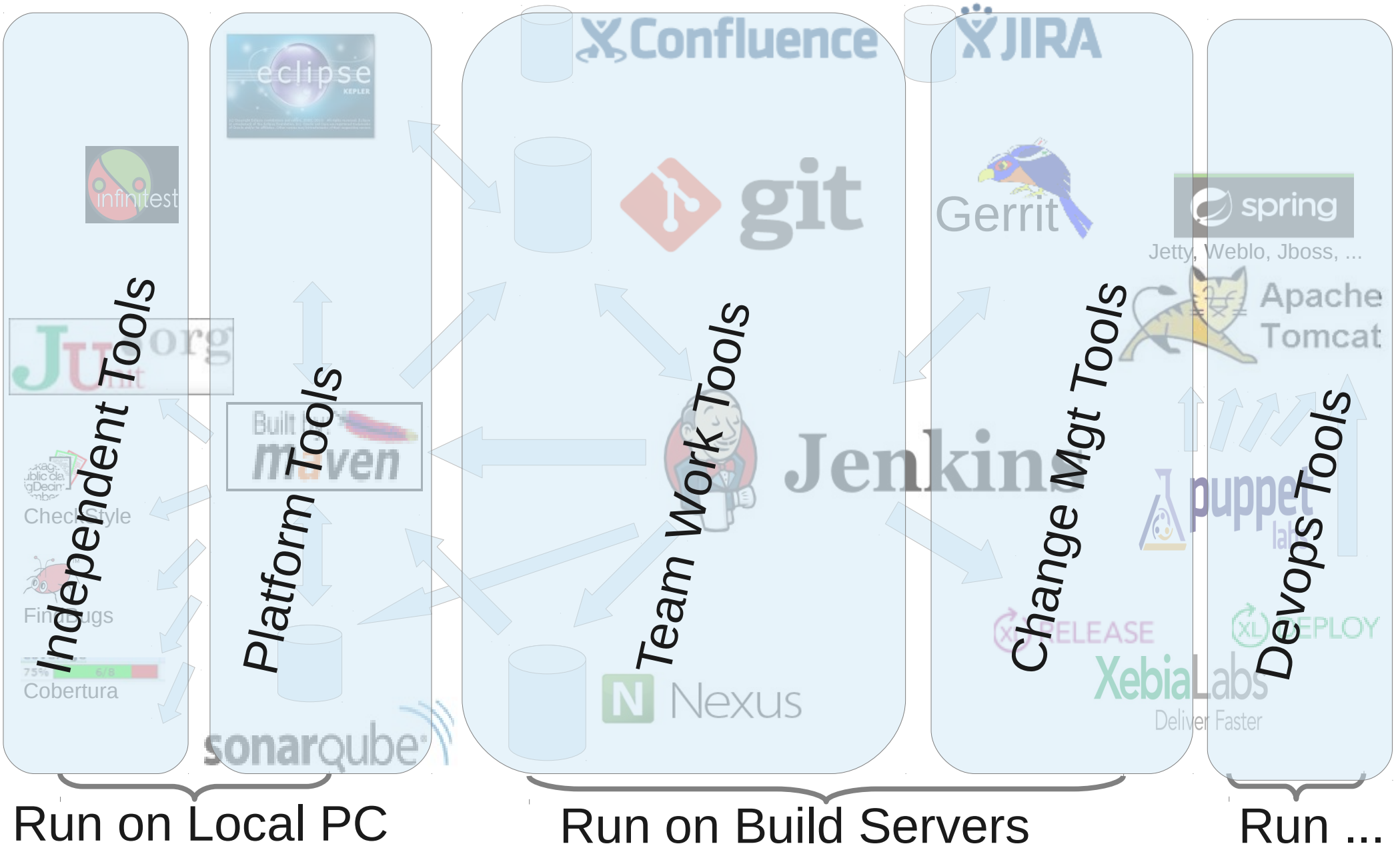
DEPLOY

XebiaLabs

Deliver Faster

pitest.org

Divide By Run Location / Category



Quizz ...
What's Common ?
(& Who are Intruders)



Confluence

JIRA

eclipse KEPLER

infinitest

git

Gerrit

spring

Jetty, Weblo, Jboss, ...

Apache Tomcat

JUnit.org

Built by: maven

Jenkins

puppet labs

CheckStyle

FindBugs

75% 6/8
Cobertura

sonarqube

Nexus

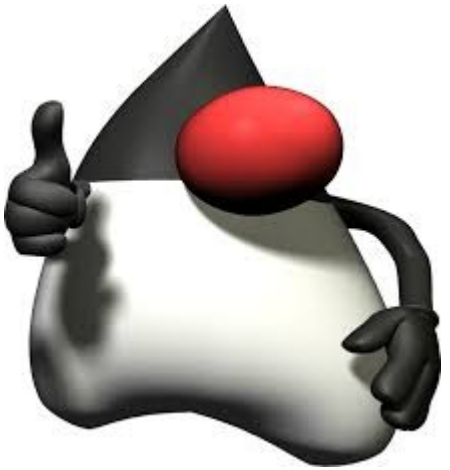
XL RELEASE

XL DEPLOY

XebiaLabs
Deliver Faster

Render unto Caesar the things...





The intruder were..



But Jgit is here :





And no real equivalent !
(glu, smartfrog??)

Commons Availability ?



eclipse KEPLER

Confluence

JIRA

infinitest

git

Gerrit

spring

Jetty, Weblo, Jboss, ...

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CheckStyle

FindBugs

75% 6/8
Cobertura

pitest.org

sonarqube

Nexus

RELEASE

DEPLOY

XebiaLabs
Deliver Faster

Open Source - Free - $ ...







Except Almost Free:

Confluence JIRA — Free Use for Open Source Projects

sonarqube Nexus — Free with $$ Support Non free Features

Except:

RELEASE DEPLOY XebiaLabs Deliver Faster — No Source & Not Free !

Tools / Langage Library ?



eclipse KEPLER

Confluence

JIRA

infinitest

git

Gerrit

spring

Jetty, Weblo, Jboss, ...

Apache Tomcat

JUnit.org

Built by: maven

Jenkins

puppet labs

CheckStyle

FindBugs

75% 6/8

Cobertura

sonarqube

Nexus

RELEASE

DEPLOY

XebiaLabs

Deliver Faster

All Tools “Agnostic”
… except these Libraries



From Beginning
(TDD Test)

JUnit.org

Java

spring

To End
(IOC, JEE App Server)

2 Java CORE Libraries
MUST-HAVE for Continuous Integration

Continuous Integration for Others ?

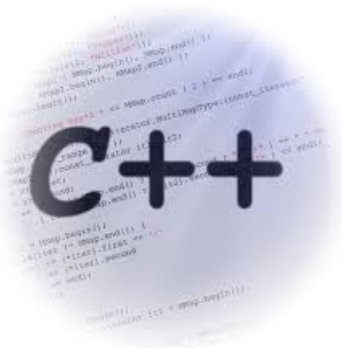








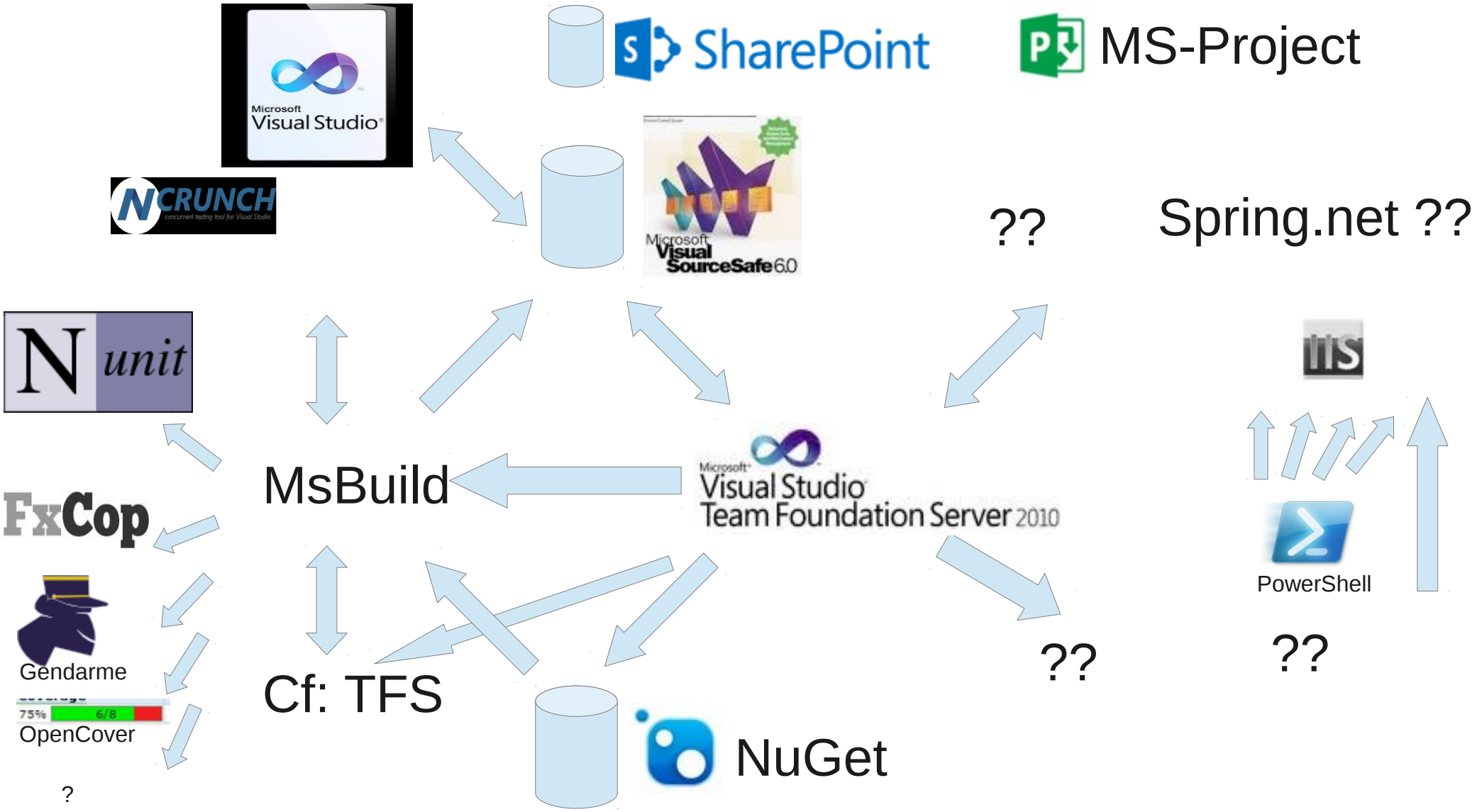




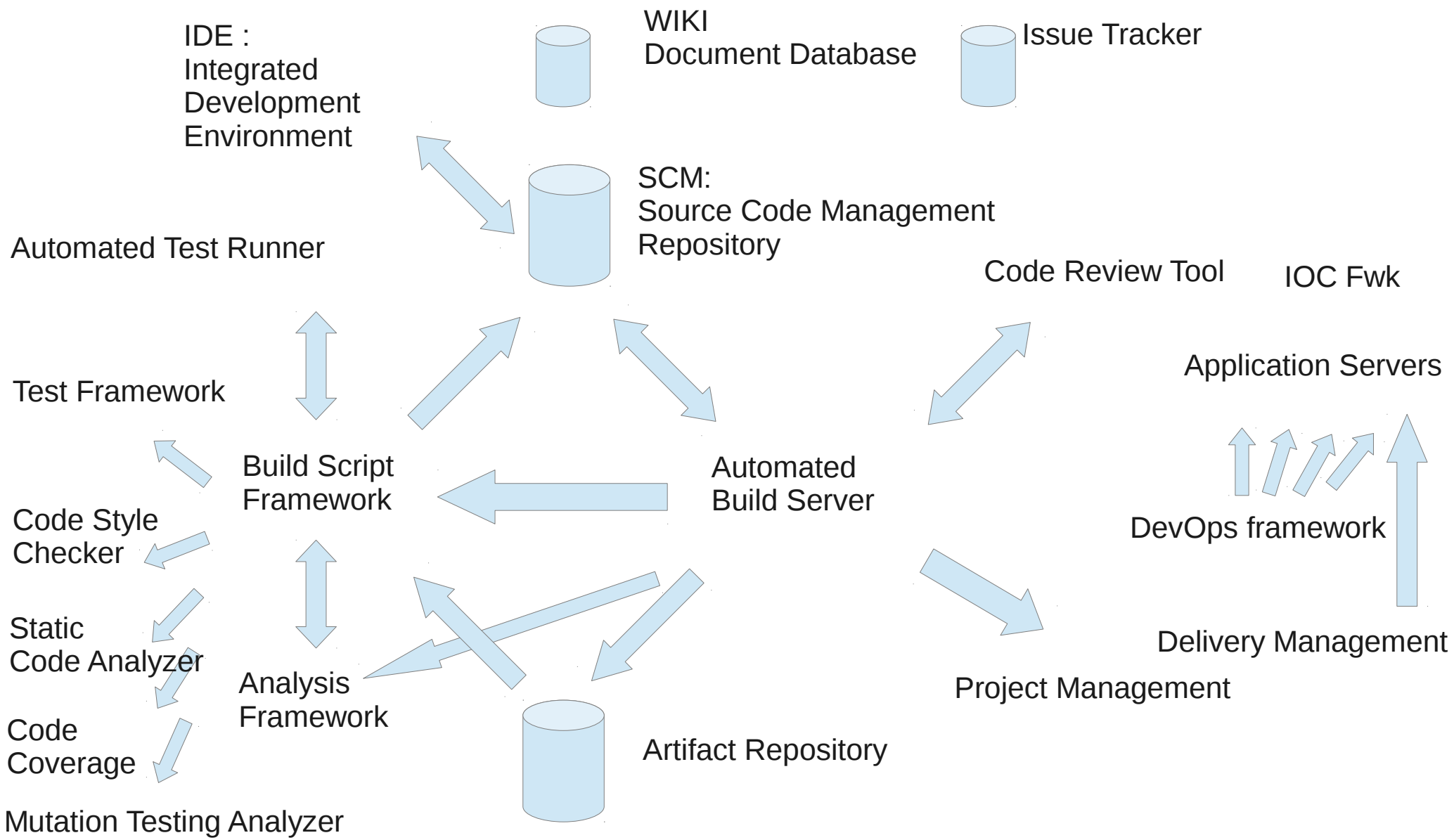




Translate => C# Micro$oft ™



Translate => Generic Name



Pause Before ...



Start Simple, and TDD Test First !



TDD

Test Driven Development

= Test First … => Test









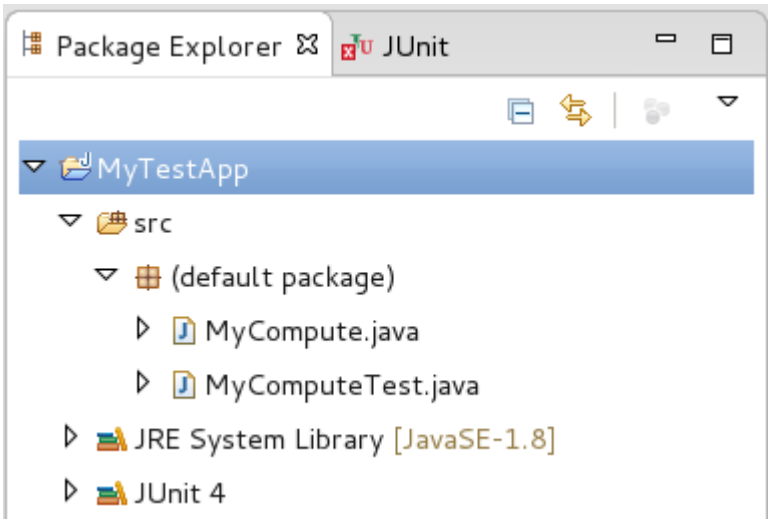






Get Hands Dirty

Create Simple Project ...




```
public class MyCompute {


	public int compute(int param1, int param2) {
		if (param2 == -2) throw new IllegalStateException();
		return param1 + ((param2 > 0)? param2 : 0);
	}

}
```

```
import org.junit.Test;

public class MyComputeTest {

	private MyCompute tested = new MyCompute();

	@Test
	public void test_1_0_equal_1() {
		// Prepare
		int param1 = 1, param2 = 0;
		// Perform
		int res = tested.compute(param1, param2);
		// Post-Check
		Assert.assertEquals(1, res);
	}

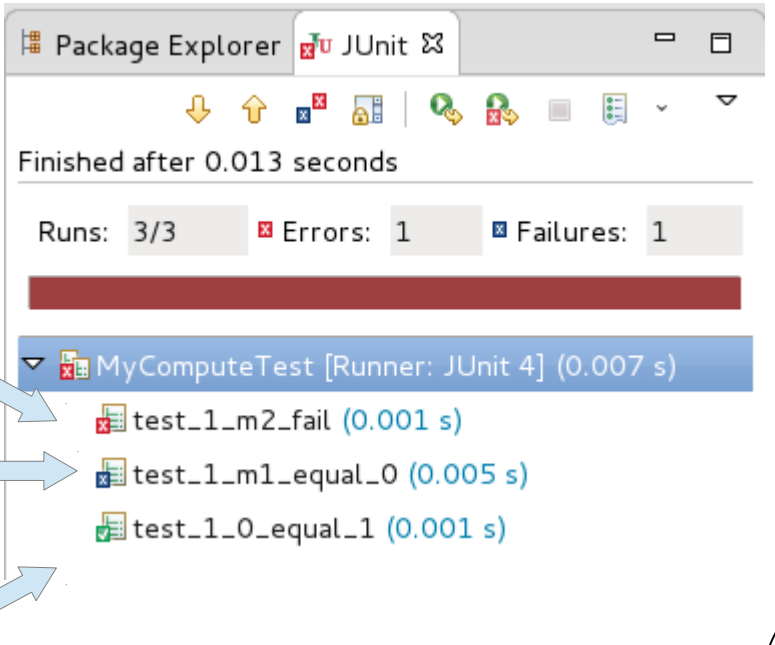
	@Test
	public void test_1_m1_equal_0() {
		// Prepare
		int param1 = 1, param2 = -1;
		// Perform
		int res = tested.compute(param1, param2);
		// Post-Check
		Assert.assertEquals(0, res);
	}

	@Test //(expected=IllegalStateException.class)
	public void test_1_m2_fail() {
		// Prepare
		int param1 = 1, param2 = -2;

		// Perform
		tested.compute(param1, param2);
	}
}
```

Running Junit Tests





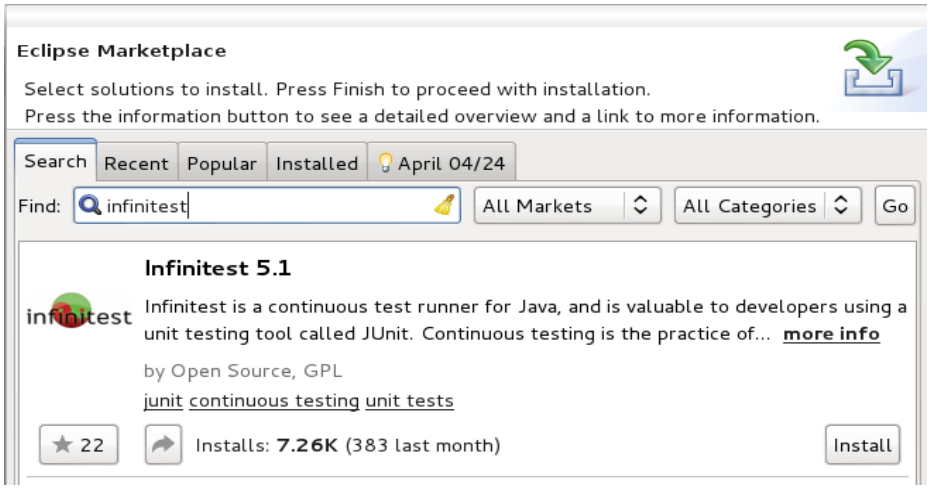
Crashed with Exception

Test Assertion Failure
expected XX, got YY

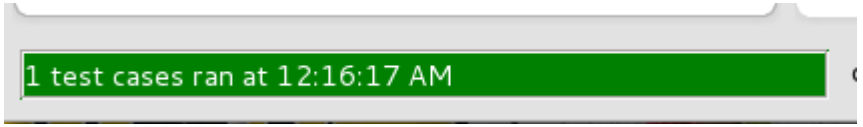
Test finished OK

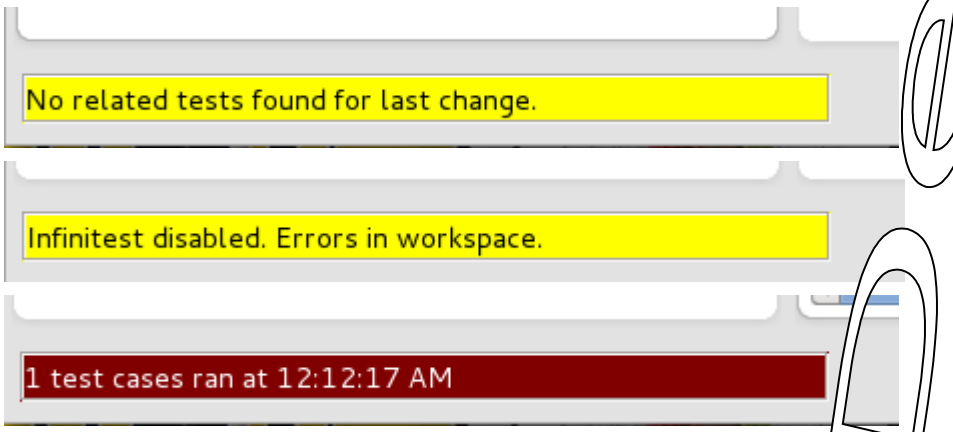
InfiniTest
Background Test Runner

Install



Then Show Running (Bottom Left)

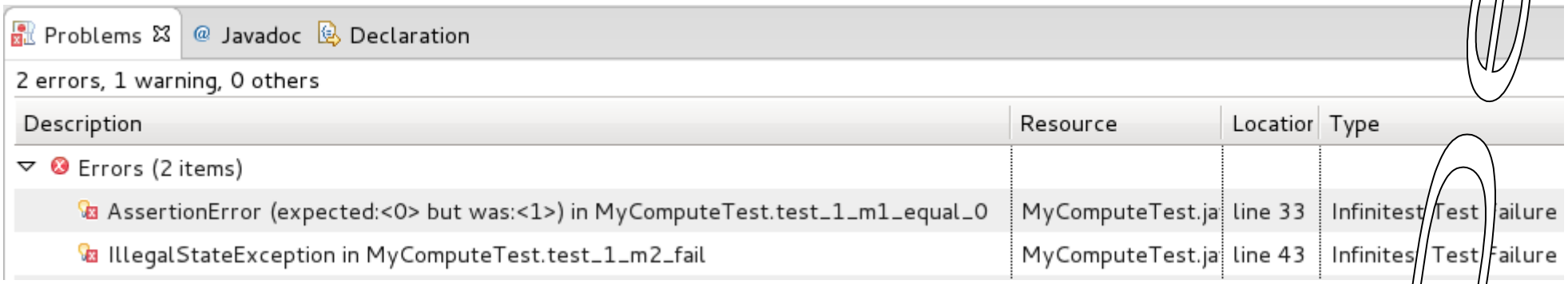




InfiniTest Markers

When Error => Shown as “Compile Error” Marker





Victory Celebration Pause
TDD in Eclipse



- [x] Develop TDD FAST
  on Local PC
  in Eclipse + Eclipse Junit + Eclipse InfiniTest

- [ ] What's next ?

- [x] Share on other PC … not having Eclipse ?

Welcome Maven pom.xml



Project POM

```
<project xmlns="http://maven.apache.org/POM/4.0.0" xmlns:xsi="http://www.w3.org
  xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org
  <modelVersion>4.0.0</modelVersion>

  <groupId>com.mycompany.app</groupId>
  <artifactId>my-app</artifactId>
  <version>1.0-SNAPSHOT</version>
  <packaging>jar</packaging>

  <name>Maven Quick Start Archetype</name>
  <url>http://maven.apache.org</url>

  <dependencies>
    <dependency>
      <groupId>junit</groupId>
      <artifactId>junit</artifactId>
      <version>4.8.2</version>
      <scope>test</scope>
    </dependency>
  </dependencies>
</project>
```

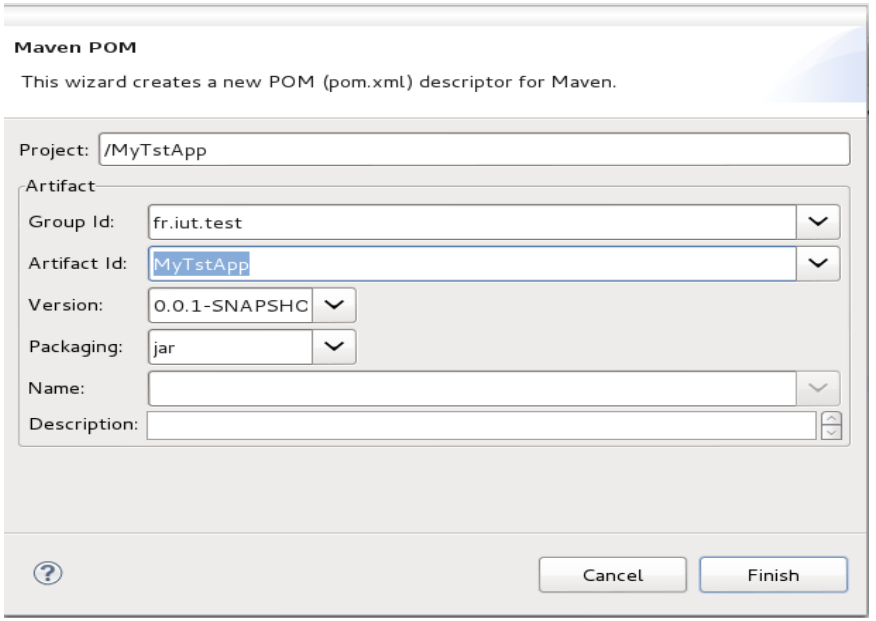
Repo

Std Layout

```
my-app
|-- pom.xml
`-- src
    |-- main
    |   `-- java
    |       `-- com
    |           `-- mycompany
    |               `-- app
    |                   `-- App.java
    `-- test
        `-- java
            `-- com
                `-- mycompany
                    `-- app
                        `-- AppTest.java
```

Get Hands Dirty
Convert to Maven - from Eclipse

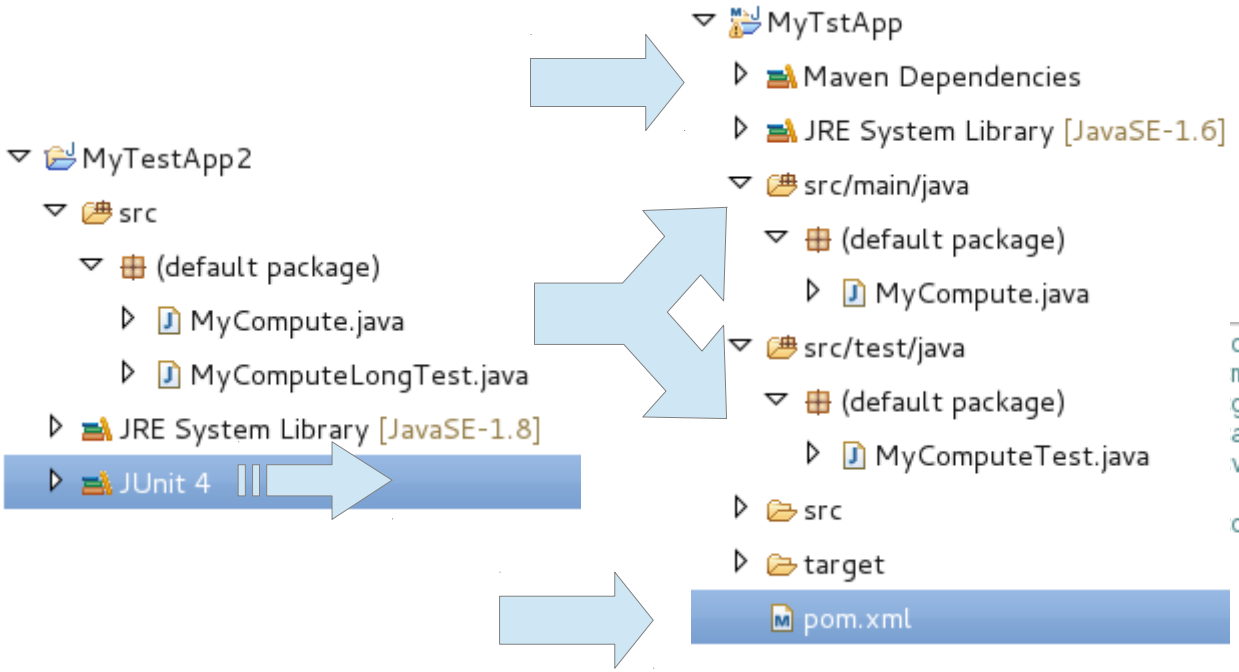






```
<project xmlns="http://maven.apache.org/POM/4.0.0" xmlr
  <modelVersion>4.0.0</modelVersion>
  <groupId>fr.iut.test</groupId>
  <artifactId>MyTstApp</artifactId>
  <version>0.0.1-SNAPSHOT</version>
  <build>
    <sourceDirectory>src</sourceDirectory>
    <plugins>
      <plugin>
        <artifactId>maven-compiler-plugin</artifactId>
        <version>3.1</version>
        <configuration>
          <source>1.6</source>
          <target>1.6</target>
        </configuration>
      </plugin>
    </plugins>
  </build>
</project>
```

Convert … src/{main,test}/java + Junit Dependency




```
oject xmlns="http://maven.apache.org/POM/4.0.0" xmlr
modelVersion>4.0.0</modelVersion>
groupId>fr.iut.test</groupId>
artifactId>MyTstApp</artifactId>
version>0.0.1-SNAPSHOT</version>

dependencies>
 <dependency>
     <groupId>junit</groupId>
     <artifactId>junit</artifactId>
     <version>4.11</version>
 </dependency>
</dependencies>

<build>
  <plugins>
    <plugin>
      <artifactId>maven-compiler-plugin</artifactId>
      <version>3.1</version>
      <configuration>
        <source>1.6</source>
        <target>1.6</target>
      </configuration>
    </plugin>
  </plugins>
</build>
</project>
```

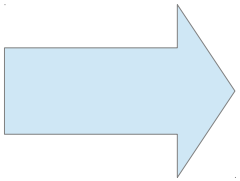
Build With Maven

Build the Project

```
mvn package
```

The command line will print out various actions, and end with the following:

```
 ...
[INFO] ------------------------------------------------------------
[INFO] BUILD SUCCESSFUL
[INFO] ------------------------------------------------------------
[INFO] Total time: 2 seconds
[INFO] Finished at: Thu Jul 07 21:34:52 CEST 2011
[INFO] Final Memory: 3M/6M
[INFO] ------------------------------------------------------------
```



Compile + test + jar …

Cf LifeCycle / Plugins Mojo

Running Maven Tools

Maven Phases

Although hardly a comprehensive list, the

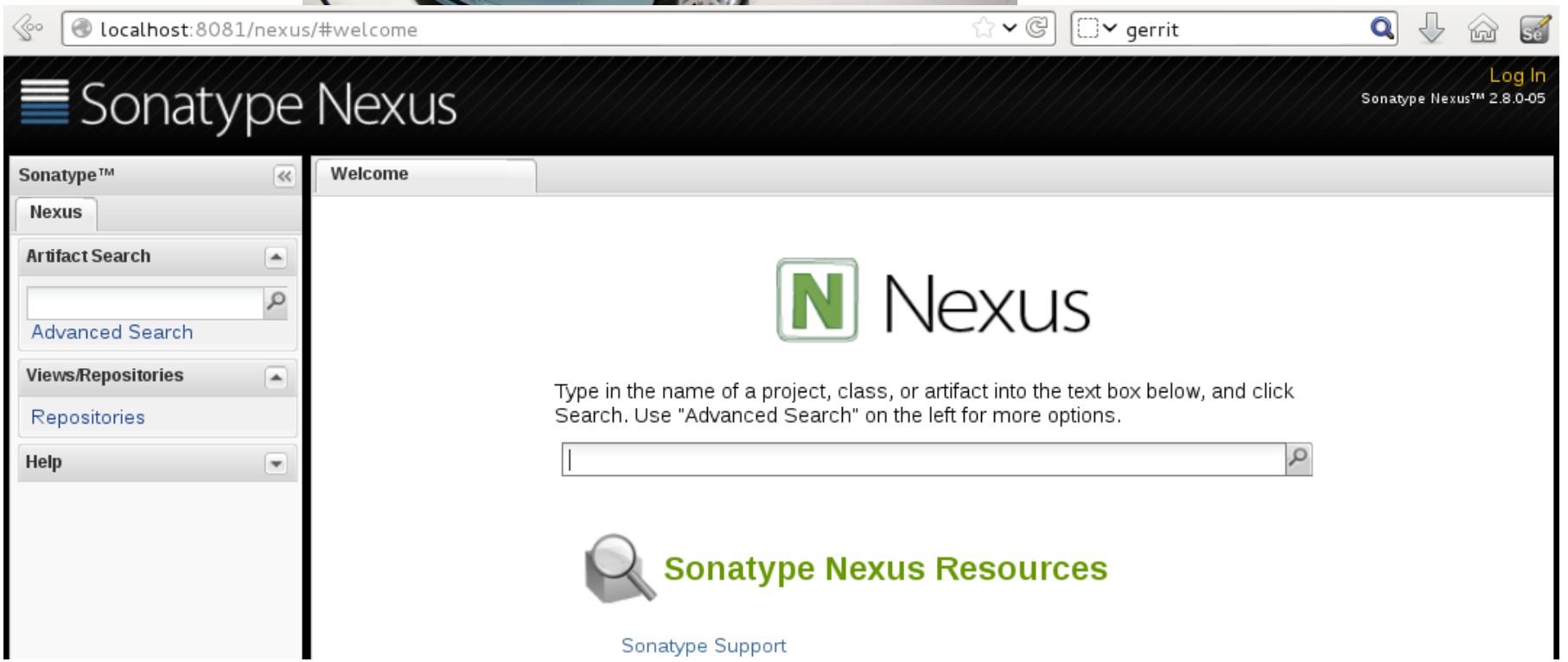
- **validate**: validate the project is c rrect a
- **compile**: compile the source cod of the
- **test**: test the compiled source c sing code be packaged or deployed
- **package**: take the compiled cod and p
- **integration-test**: process and d y the be run
- **verify**: run any checks to verify t packa
- **install**: install the package int the local
- **deploy**: done in an integration release sharing with other developers and proje

There are two other Maven lifecycl s emote be

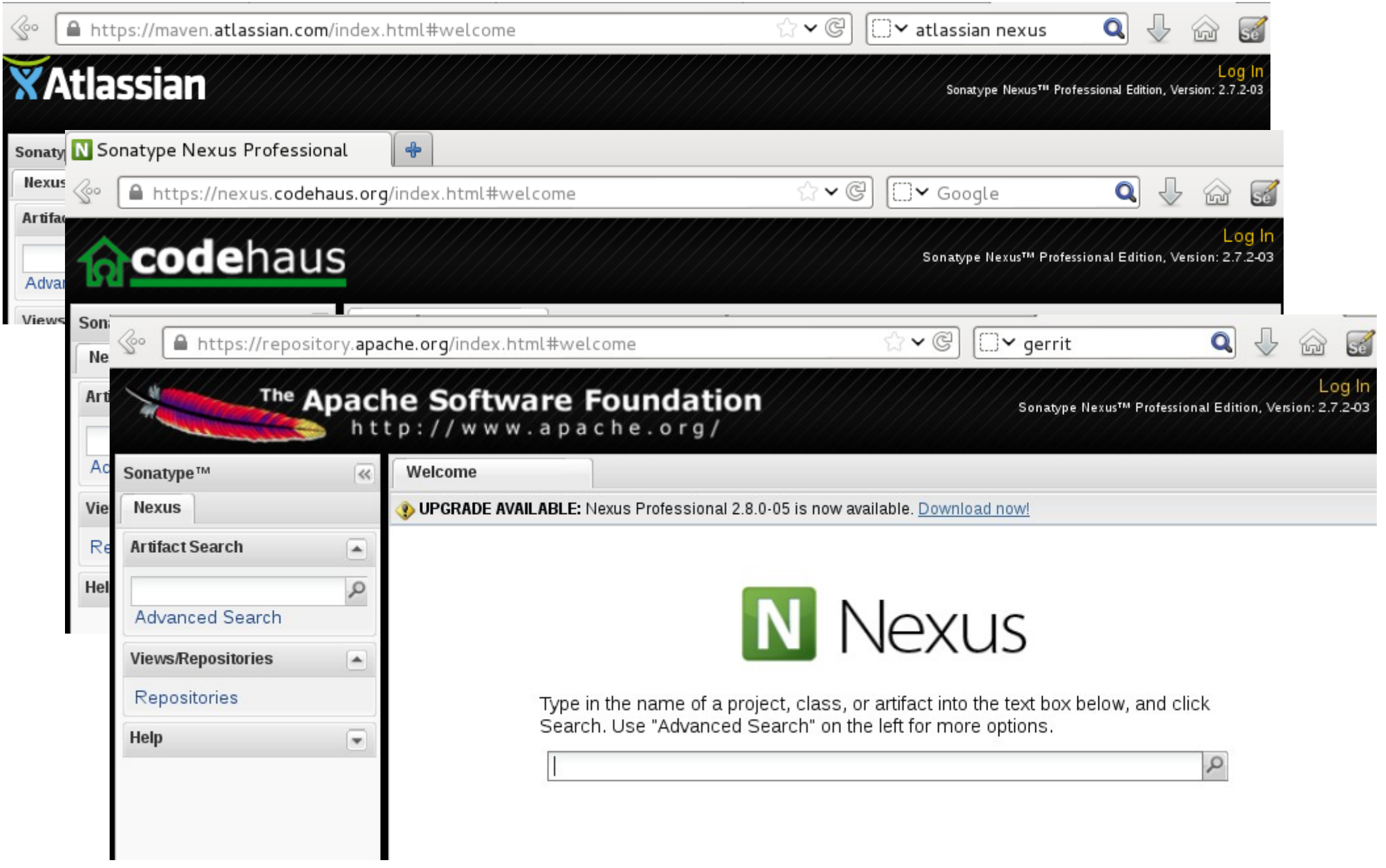
- **clean**: cleans up artifacts reate by prio
- **site**: generates site docu enta on for th

Download Dependencies ?
.. cf Nexus Repository

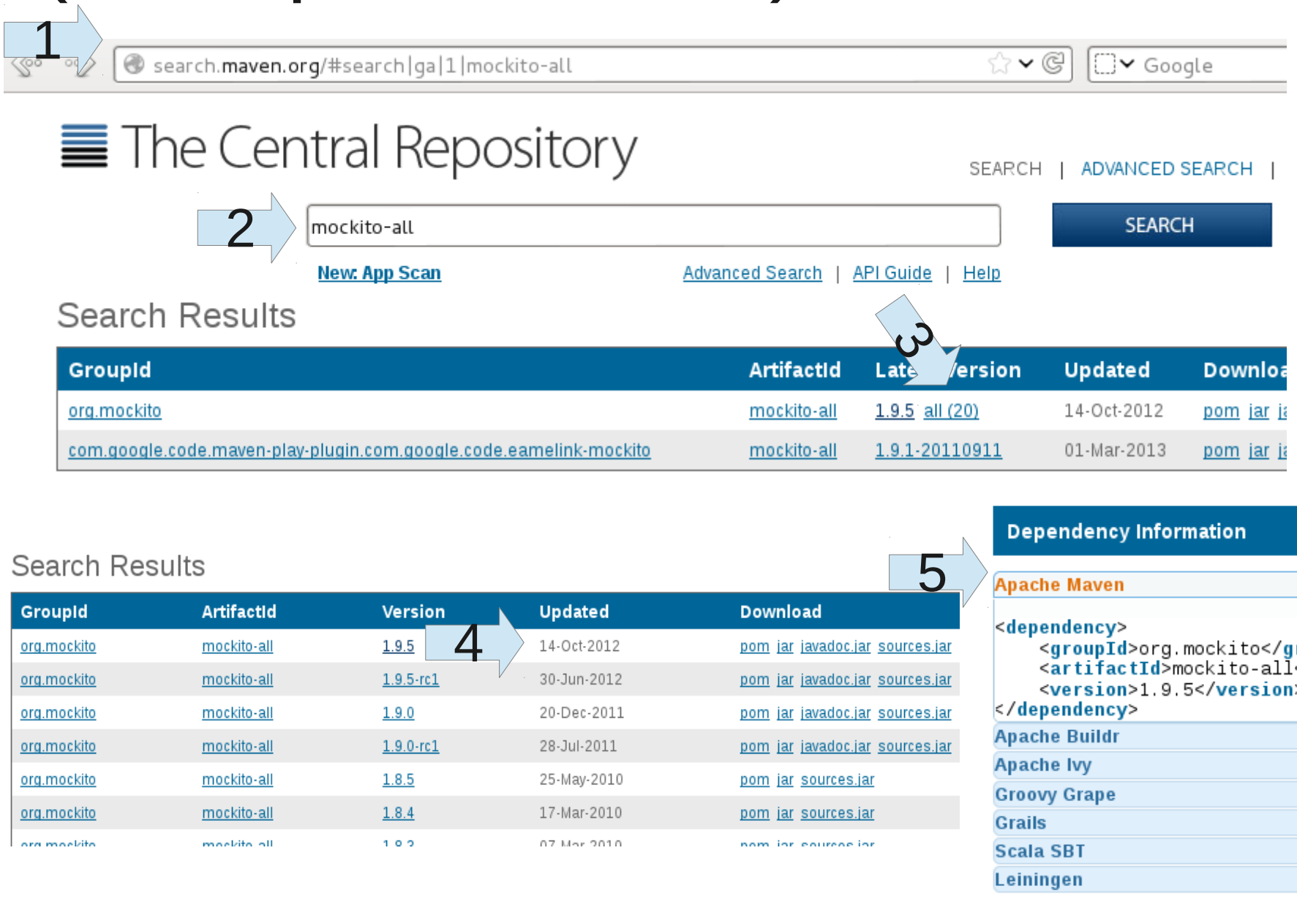




Public Repositories on Internet

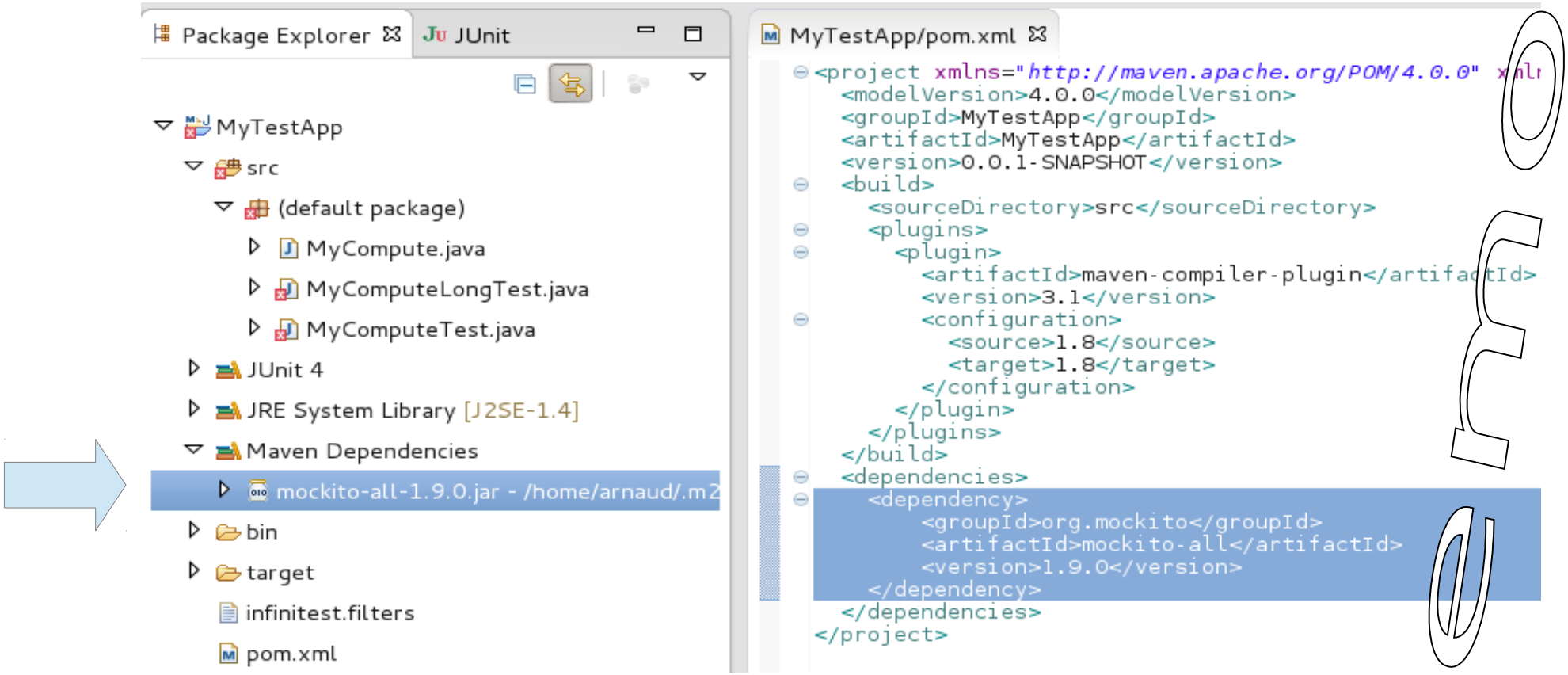


Search Maven Dependency (Example : Mockito)



Add Dependency in Pom.xml

To Download … simply wait eclipse auto-build !!



Or “mvn install”

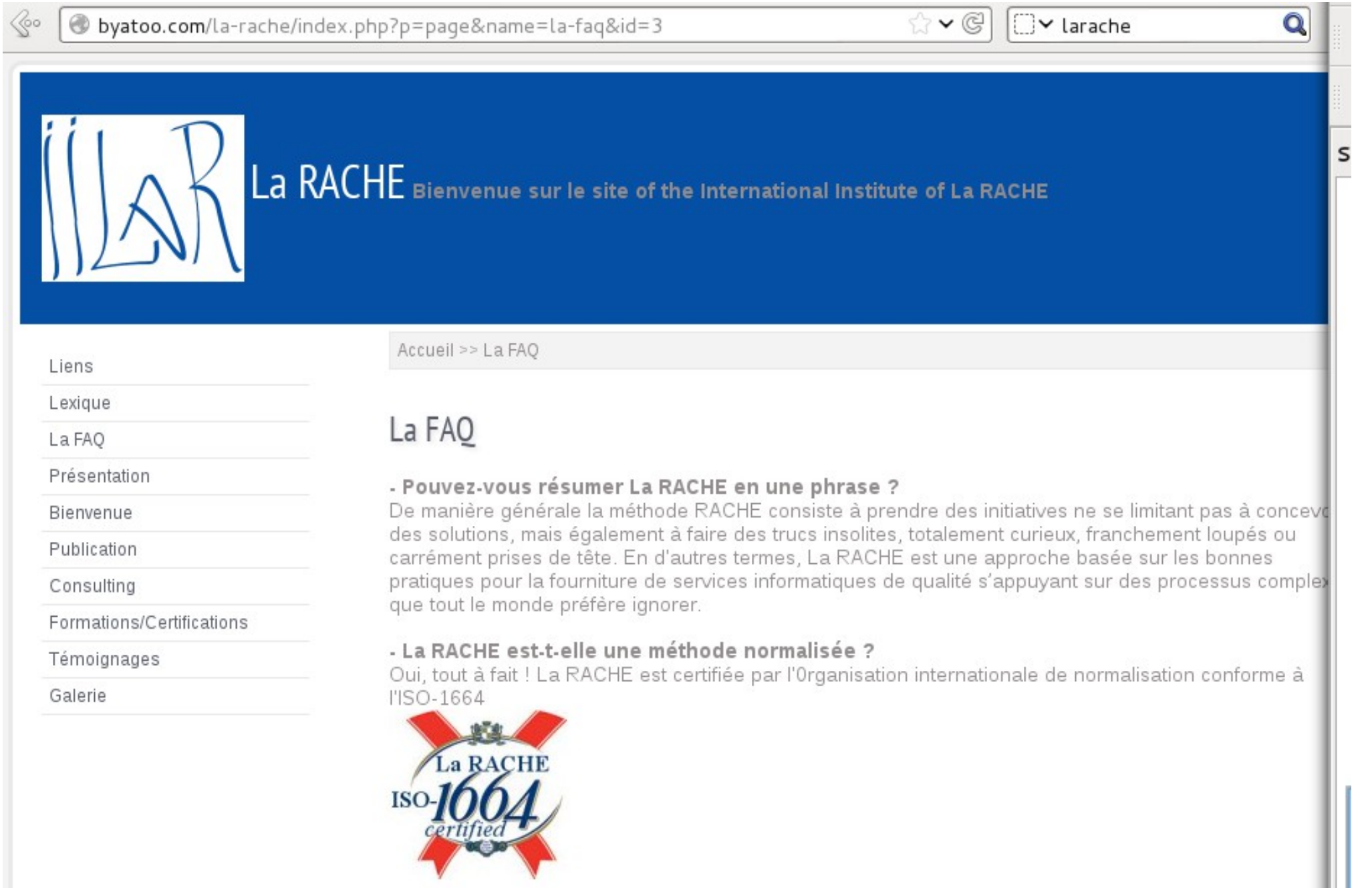
Maven

- TRULY AWESOME … Industry Standard … MUST-HAVE
- But Xml verbose...
- Deprecated Alternatives
  Compile.sh / compile.bat … with javac + jar, Makefile

  ant 

- For C# … Msbuild + nuget
- Other Modern alternatives ? gradle ivy

Build Without Maven ? ...



Victory Celebration Pause





Maven – platform independent build

Questions so far ?

Victory Celebration Pause



- [x] Maven – platform independent build
- [ ] What's next ?

- [x] Backup on usb key?
- [x] Share / email changes to colleagues ?
- [x] Apply changes

Questions so far ?

GIT – Source Code Mgt

- [x] Local Dev
- [ ] Team Dev

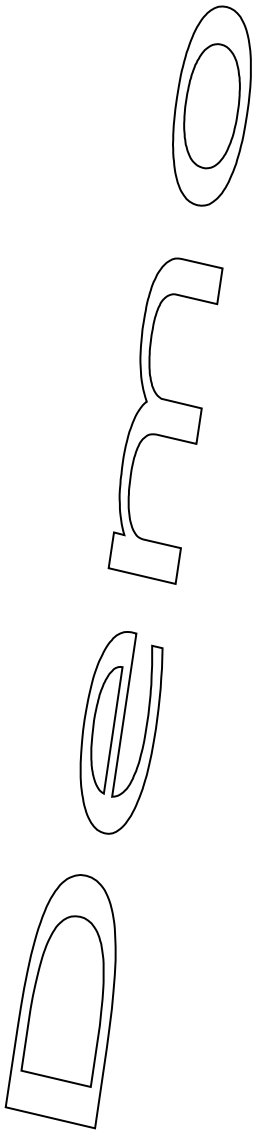
- [x] Backup on usb key?
- [x] Share / email changes to colleagues ?
- [x] Apply changes



- [x] Git Commit
- [x] Git Push
- [x] Git Branch/Merge

1 minute GIT Demo

```
$ git init
Initialized empty Git repository in /mnt/a_1tera2/homeData/arnaud/
/continous-int/demo-dir/my-app/.git/
$
$ git st
# On branch master
#
# Initial commit
#
# Untracked files:
#   (use "git add <file>..." to include in what will be committed)
#
#       .gitignore
#       pom.xml
#       src/
nothing added to commit but untracked files present (use "git add"
$
$ git add pom.xml src
$
$ git commit -m "init"
[master (root-commit) e6a19da] init
 3 files changed, 69 insertions(+)
 create mode 100644 pom.xml
 create mode 100644 src/main/java/com/mycompany/app/App.java
 create mode 100644 src/test/java/com/mycompany/app/AppTest.java
$
```



Start Git Server

1) Copy to ”Bare” repos

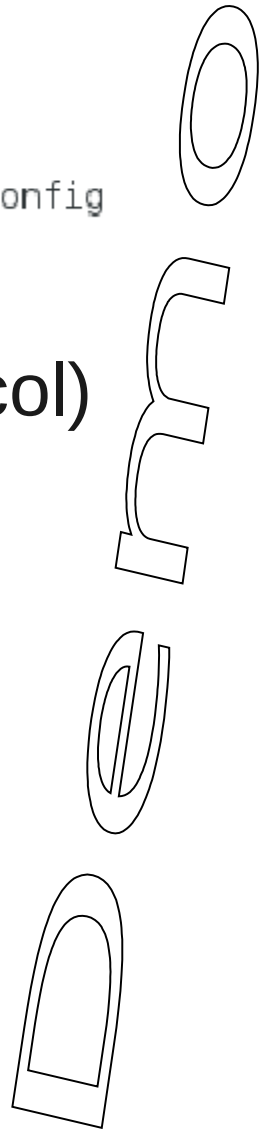
```
$ mkdir git-repos
$ cd git-repos
$ cp -rf ../testfoo-mvc/.git testfoo-mvc.git
$ cat testfoo-mvc.git/config | sed 's/bar = false/bare = true/g' > testfoo-mvc.git/config
$ touch testfoo-mvc.git/git-daemon-export-ok
```

2) Start Simplest as possible server = (git:// protocol)

```
$
$ git daemon --reuseaddr --base-path=. --export-all --enable=receive-pack
```

3) Clone on client

```
$
$ git clone git://localhost/testfoo-mvc
Cloning into 'testfoo-mvc'...
remote: Counting objects: 132, done.
remote: Compressing objects: 100% (119/119), done.
remote: Total 132 (delta 30), reused 0 (delta 0)
Receiving objects: 100% (132/132), 75.00 KiB, done.
Resolving deltas: 100% (30/30), done.
$
```



Commit – Push
Pull (fetch+merge)

1) Local commit

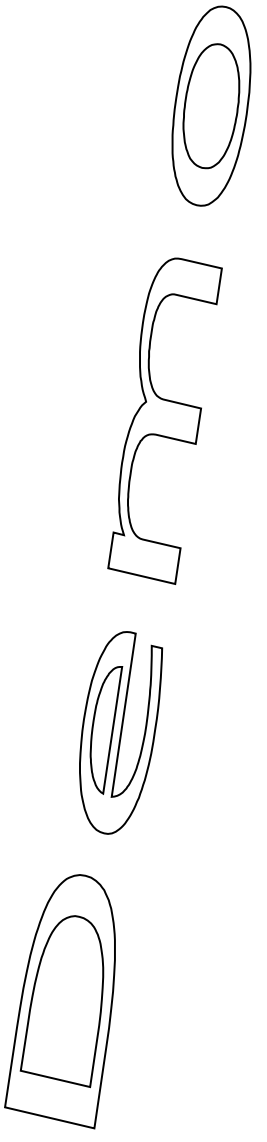
```
$ git add pom.xml
$ git commit -m "added comment"
[master b3dfcec] added comment
 1 file changed, 1 insertion(+)
```

2) Push to Git server

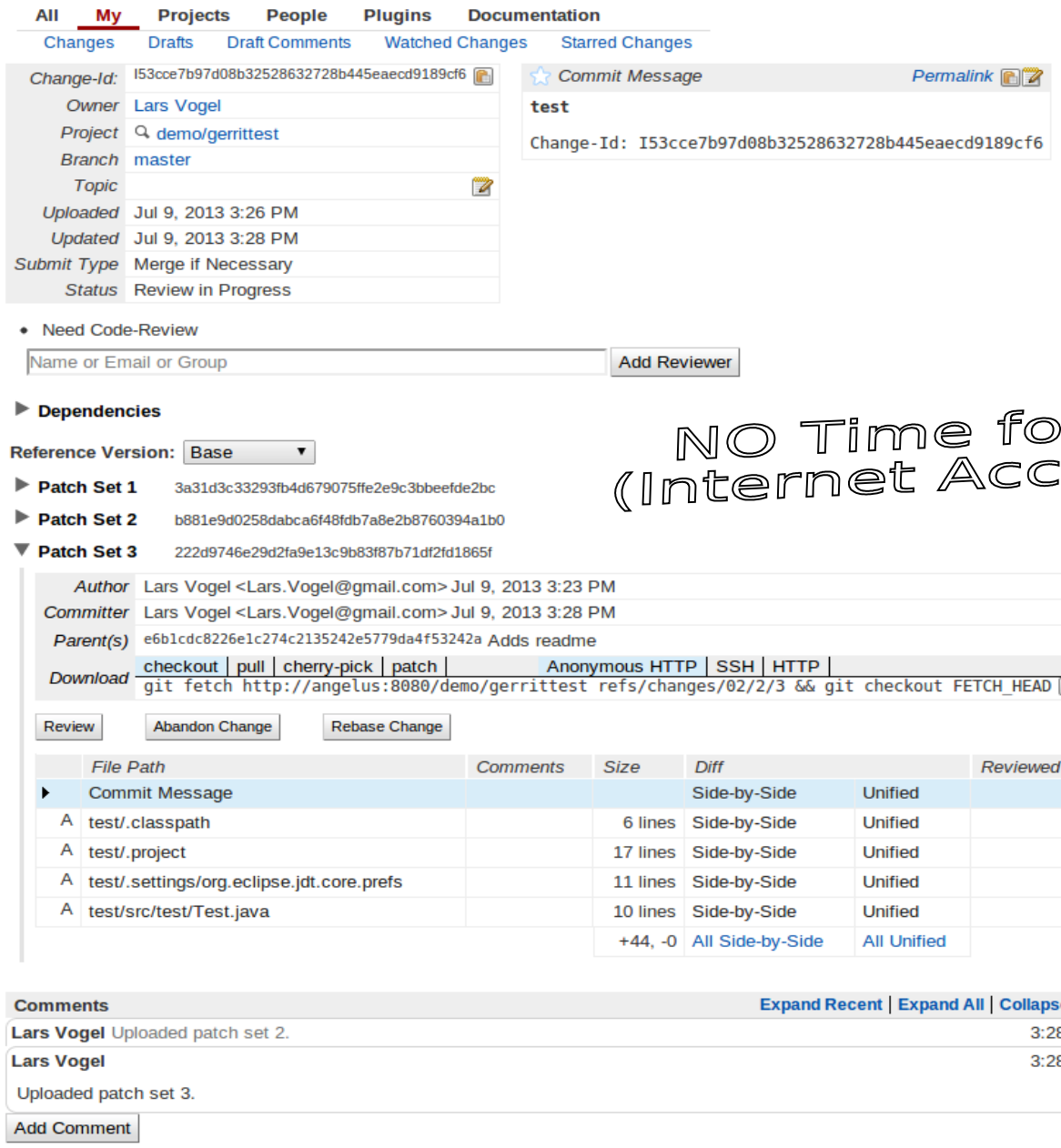
```
$ git push origin
Counting objects: 5, done.
Delta compression using up to 4 threads.
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 294 bytes, done.
Total 3 (delta 2), reused 0 (delta 0)
To git://localhost/testfoo-mvc
   4b3a983..b3dfcec  master -> master
$
```

3) Pull from other

```
$ git pull origin master
From git://localhost/testfoo-mvc
 * branch            master     -> FETCH_HEAD
Updating d2fa761..4b3a983
Fast-forward
 pom.xml |    5 +++++
 1 file changed, 5 insertions(+)
$
```



Gerrit Code Review



NO Time for Demo Here !!
(Internet Access – OpenID) ??

Victory Celebration Pause



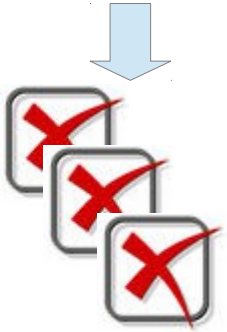
- [x] Maven – platform independent build
- [x] GIT Source Code Management

Questions so far ?

Victory Celebration Pause



- [x] Maven – platform independent build
- [x] GIT Source Code Management
- [ ] What's next ?



Repeat Compile on every Commit

...

...

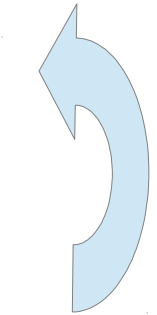
Local Dev Loop
FAST Part of Continuous Integration



Develop

Build + Test

Refactor



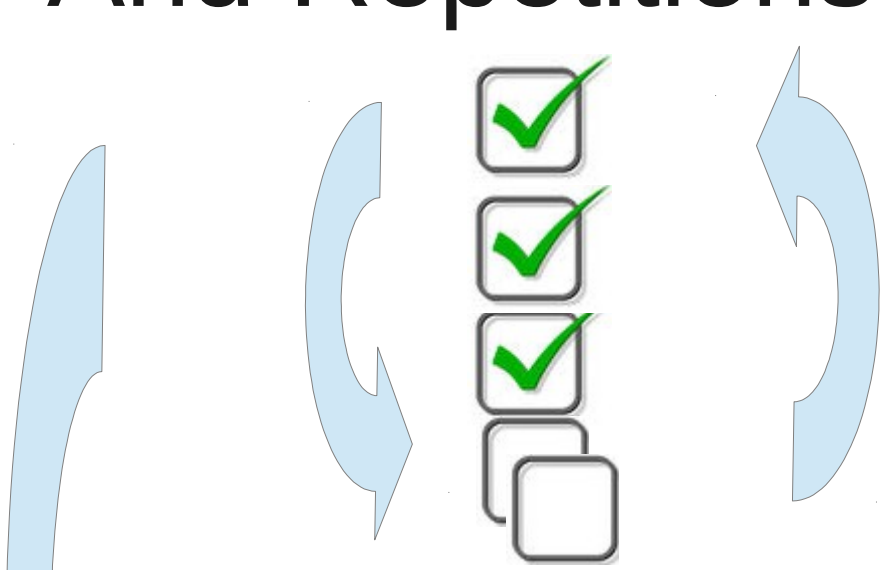
(Agile Pulse)



GIT Commit + Push

Pomodoro & Coffe-Time Break

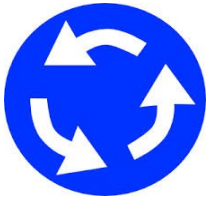
Slow Parts Hidden ?
And Repetitions ?



- [x] GIT Commit + Push
- [x] Compile scripts, post-build scripts
- [x] Build / Test history
- [x] Email, Publish, Reports ...
- [x] Deliver To UAT..Prod from my PC



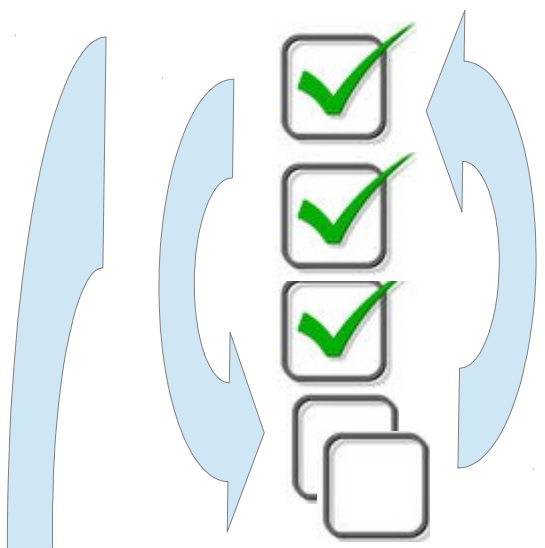
Continuously … 1000x



Type “mvn install” + email
+ archive tests + archive build
+ copy + start/stop... + …



Continuous Integration Requirements : FASTER outer Loop



Develop

Build + Test

Refactor (Agile Pulse)

Whatever if Automated & transparent to me ...

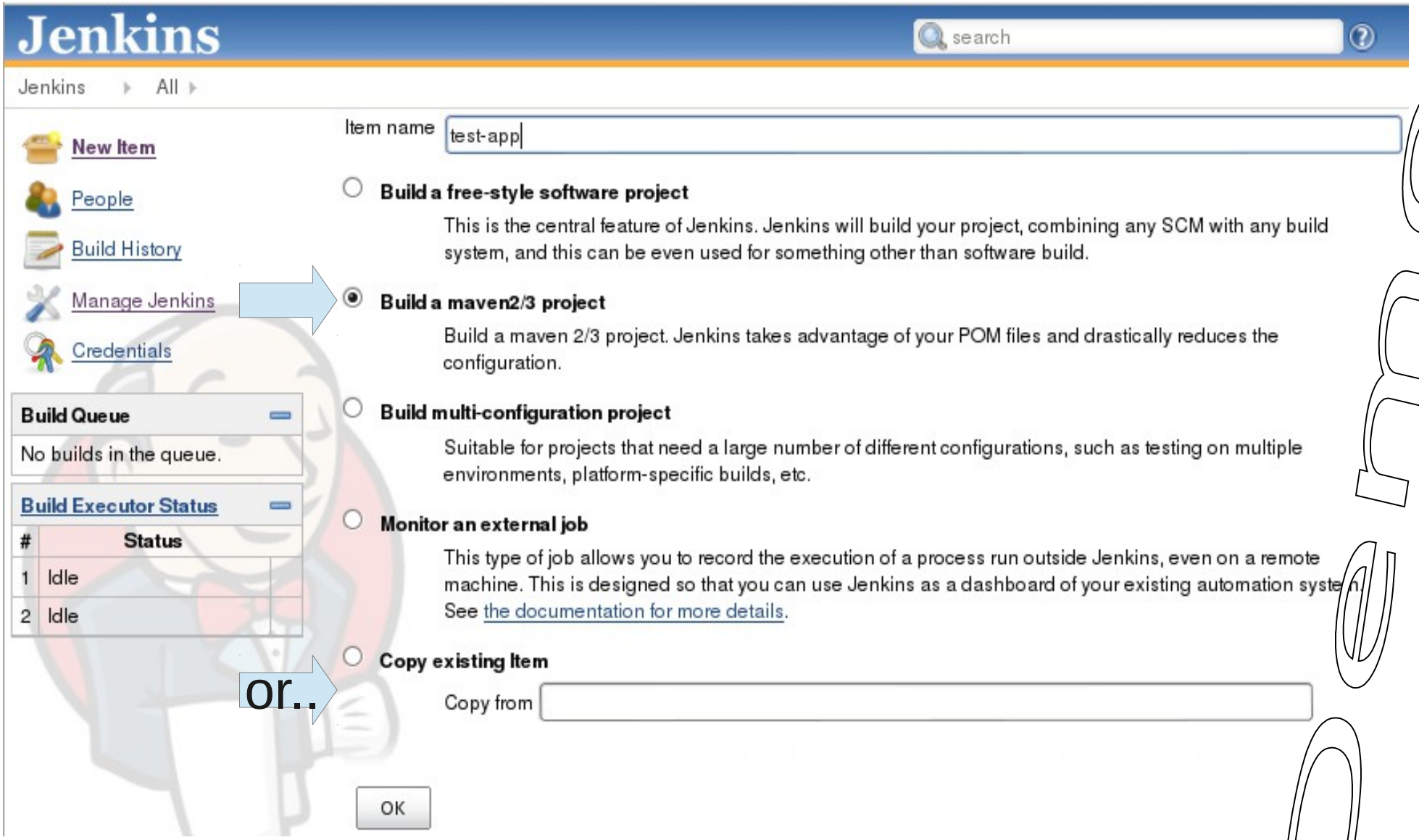


Jenkins



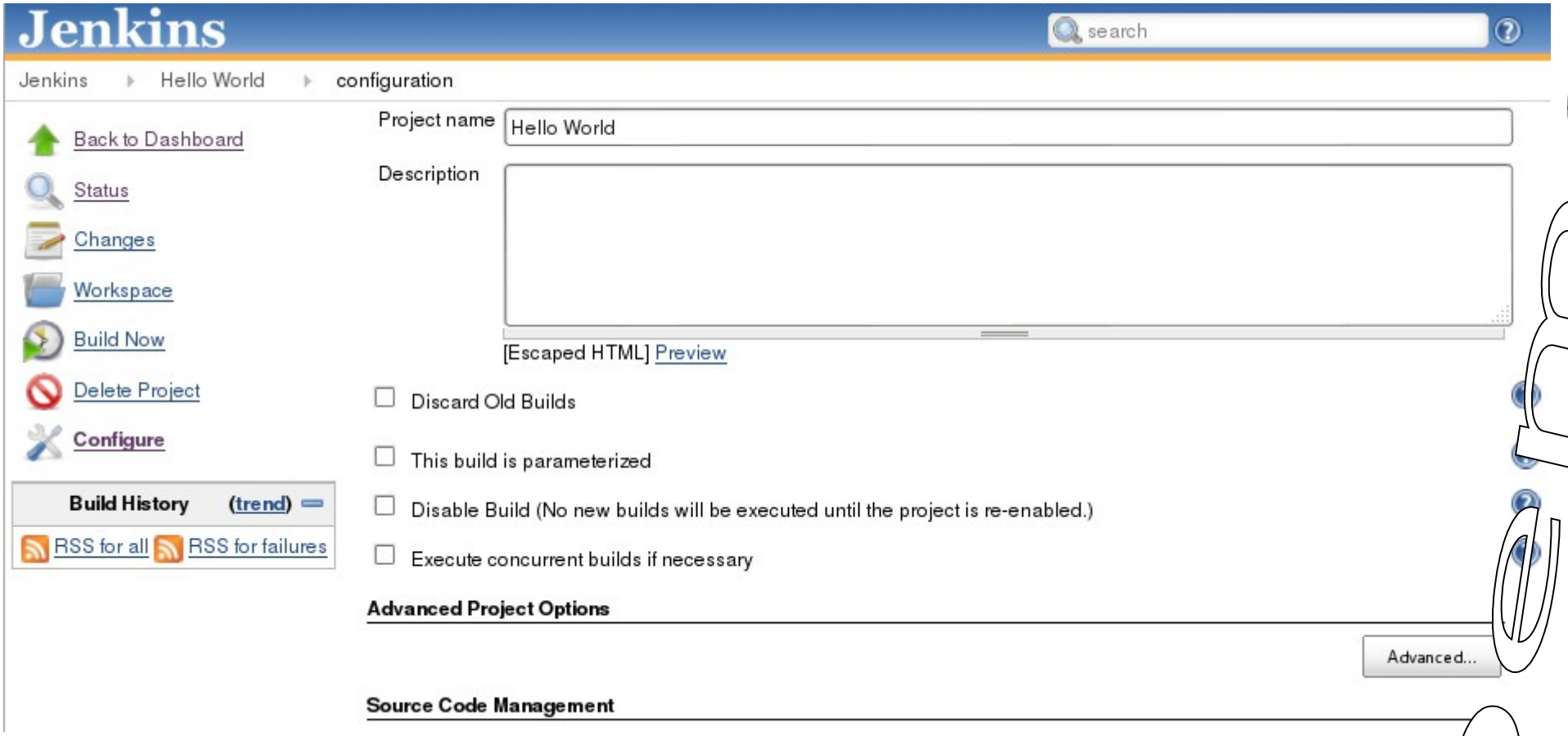
Whatever if Automated & transparent to others

New Jenkins Maven Job



Create First Job (1/3)





… fill : SCM + Build Trigger

Job > Conf > SCM



Jenkins

Source Code Management

- ○ CVS
- ○ CVS Projectset
- ○ File System
- ◉ Git

Repositories

Repository URL: git://localhost/testfoo-mvc

Credentials: - none -

Add

Advanced...

Add Repository | Delete Repository

Branches to build

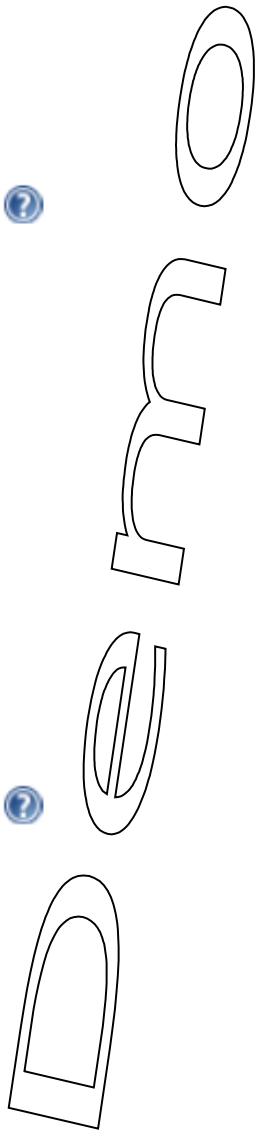
Branch Specifier (blank for 'any'): */master

Add Branch | Delete Branch

Repository browser: (Auto)

Additional Behaviours: Add

- ○ None
- ○ Subversion



Job > Conf > Build Trigger

Jenkins

Build Triggers

- [ ] Build whenever a SNAPSHOT dependency is built
- [ ] Build after other projects are built
- [ ] Build periodically
- [ ] Gerrit event
- [x] Poll SCM

Schedule: `H/2 * * * *`

Would last have run at Wednesday, May 7, 2014 10:18:46 PM CEST; wou
2014 10:18:46 PM CEST.

Ignore post-commit hooks [ ]

Advanced Project Options

- [x] Quiet period

Quiet period: 30

Number of seconds

- [ ] Retry Count

REQUIRED for automated build on commit ...

Git Polling Log

Git Polling Log

```
Started on May 7, 2014 10:23:48 PM
Using strategy: Default
[poll] Last Built Revision: Revision 26ecaab
Done. Took 43 ms
Changes found
```

Git Polling Log

Build History (trend)

#9 (pending—In the quiet period. Expires in 29 sec)

#8 May 7, 2014 10:21:56 PM

Build History (trend)

#4 May 7, 2014 10:11:48 PM

#3 May 7, 2014 10:09:48 PM

#2 May 7, 2014 10:07:48 PM

#1 May 7, 2014 9:46:28 PM

RSS for all RSS for failures

Conf > Maven + Steps



Jenkins

Pre Steps

Add pre-build step

Build

Maven Version: maven-3.2.1

Root POM: pom.xml

Goals and options:

Advanced...

Post Steps

Run only if build succeeds | Run only if build succeeds or is unstable | Run regardless of build result

Should the post-build steps run only for successful builds, etc.

Add post-build step

Build Settings

E-mail Notification

Post-build Actions

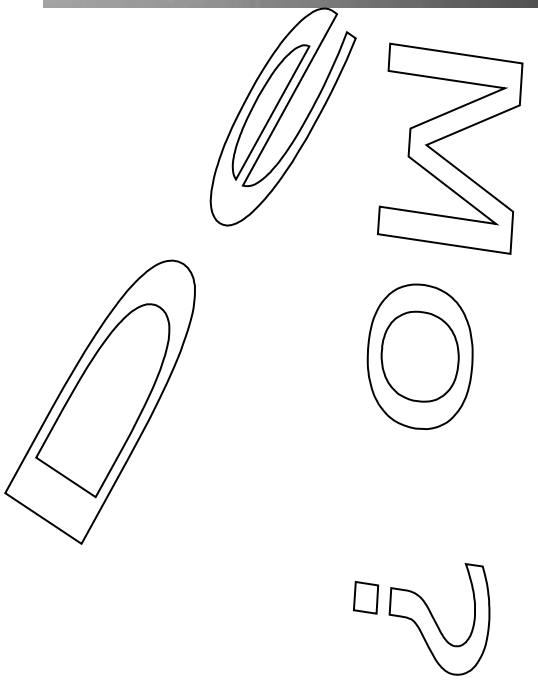
Add post-build action

Relaunch Build Now
.. or Trigger on Commit

Jenkins



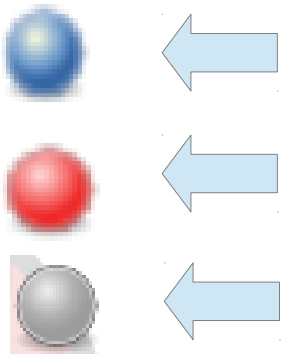




It Works ! Red → Blue





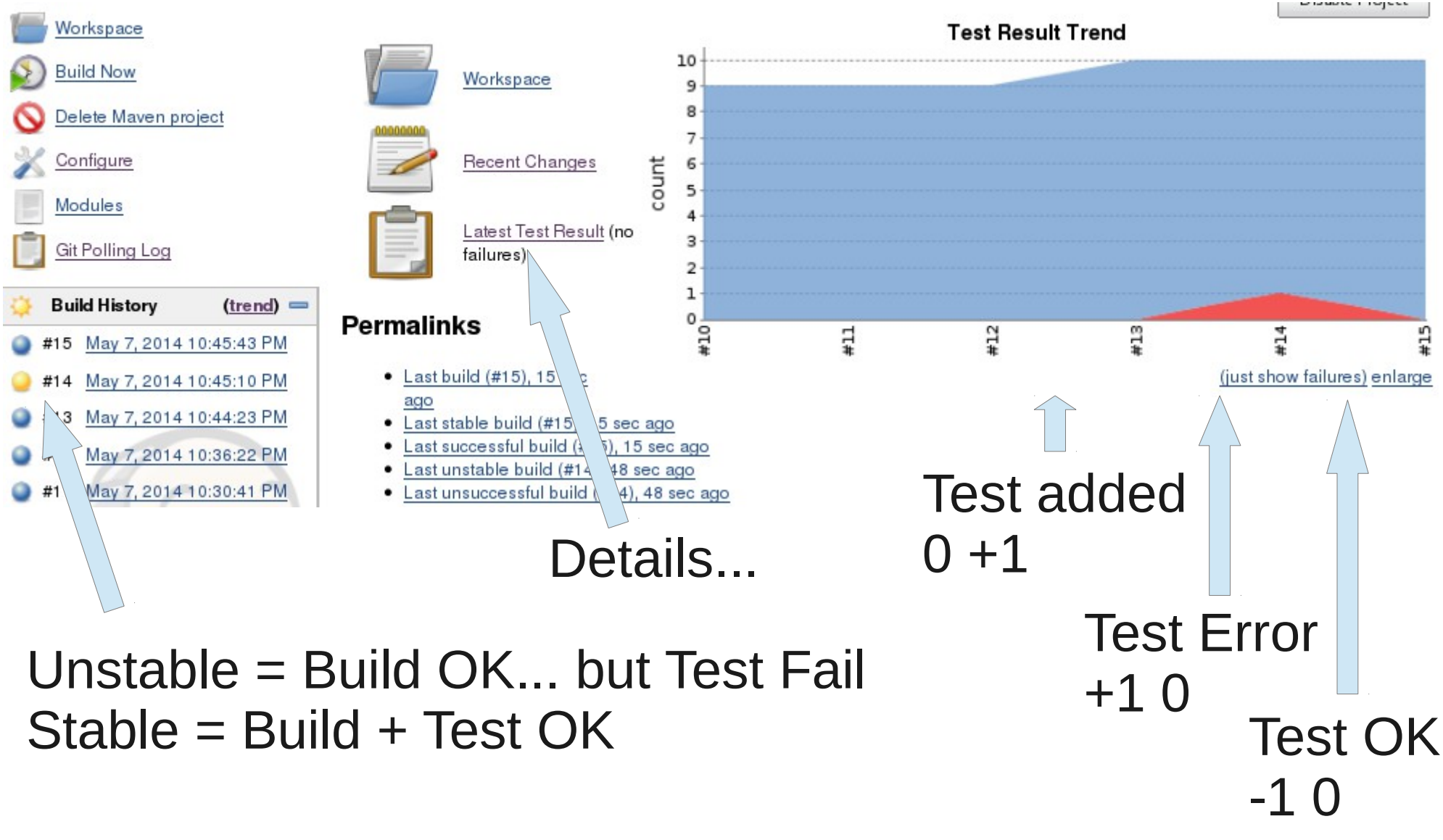


Build SUCCESS

Build FAILED

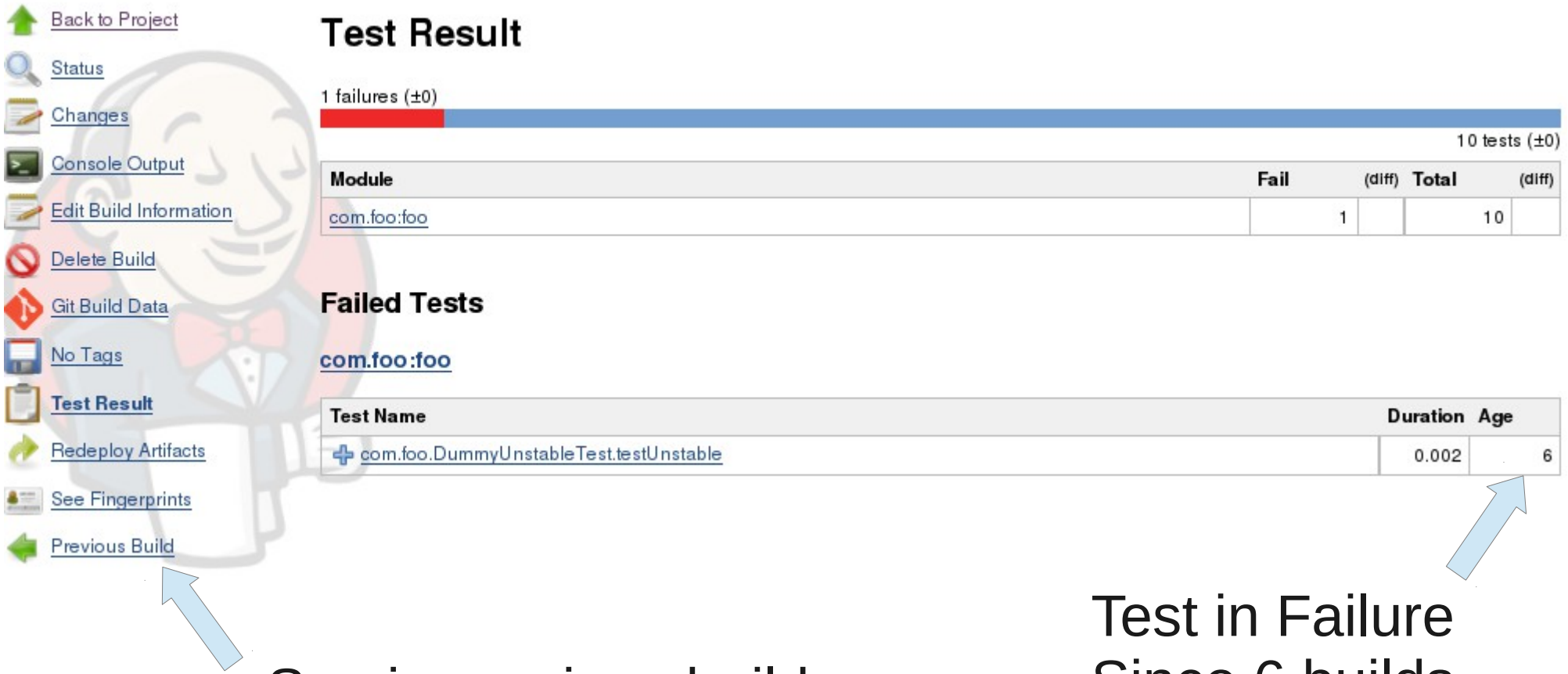
Job Disabled (view in dashboard)

Buid Result …
Junit Test Trend



Unstable = Build OK... but Test Fail
Stable = Build + Test OK

Test Result History Details



See in previous build

Test in Failure
Since 6 builds

Test Result



1 failures (+1)

10 tests (±0)

Module	Fail	(diff)	Total	(diff)
com.foo:foo	1	+1	10	

Failure = +1
this build

Failed Tests

com.foo:foo

Test Name	Duration	Age
com.foo.DummyUnstableTest.testUnstable	0.002	1

Failure detail:
Log + assert
+ exception

Age=1 :
Test is NEW Failure

Regression

com.foo.DummyUnstableTest.testUnstable

Failing for the past 1 build (Since #16)
Took 2 ms.
add description

Error Message

```
expected:<0> but was:<1>
```

Stacktrace

```
java.lang.AssertionError: expected:<0> but was:<1>
        at org.junit.Assert.fail(Assert.java:88)
        at org.junit.Assert.failNotEquals(Assert.java:743)
        at org.junit.Assert.assertEquals(Assert.java:118)
        at org.junit.Assert.assertEquals(Assert.java:555)
        at org.junit.Assert.assertEquals(Assert.java:542)
        at com.foo.DummyUnstableTest.testUnstable(DummyUnstableTest.java:19)
```

Dashboard Meteo Per Job



Last Build (binary value)

Average Recent Builds Range scale = Meteo clouds



Meteo Example

https://hudson.eclipse.org/hudson/



People
Build History
Project Relationship
Check File Fingerprint

Build Queue
No builds in the queue.

Build Executor Status
Master 0/1
Idle
Fastlane 0/2
Idle
hudson-perf1-tests 0/1
Idle
hudson-slave1 0/5
Idle
hudson-slave2 0/2
Idle
hudson-slave4 0/5
Idle
hudson-slave6 0/2
Idle
hudson-slave7(ppc64) offline
Offline
hudson-slave8(ppc64) 0/8
Idle
mac-tests 0/4
Idle
mac-tests2 0/4
Idle
windows7tests 0/1
Idle

Eclipse.org Hudson for builds

Go: Hudson Sandbox
- HIPP instances: ECF | EGit | eTrice | Gyrex | Hudson project | LinuxTools | OCL | Orion | PTP | Sapphire SWTBot | UOMo Xtext |
- Request HIPP for your project

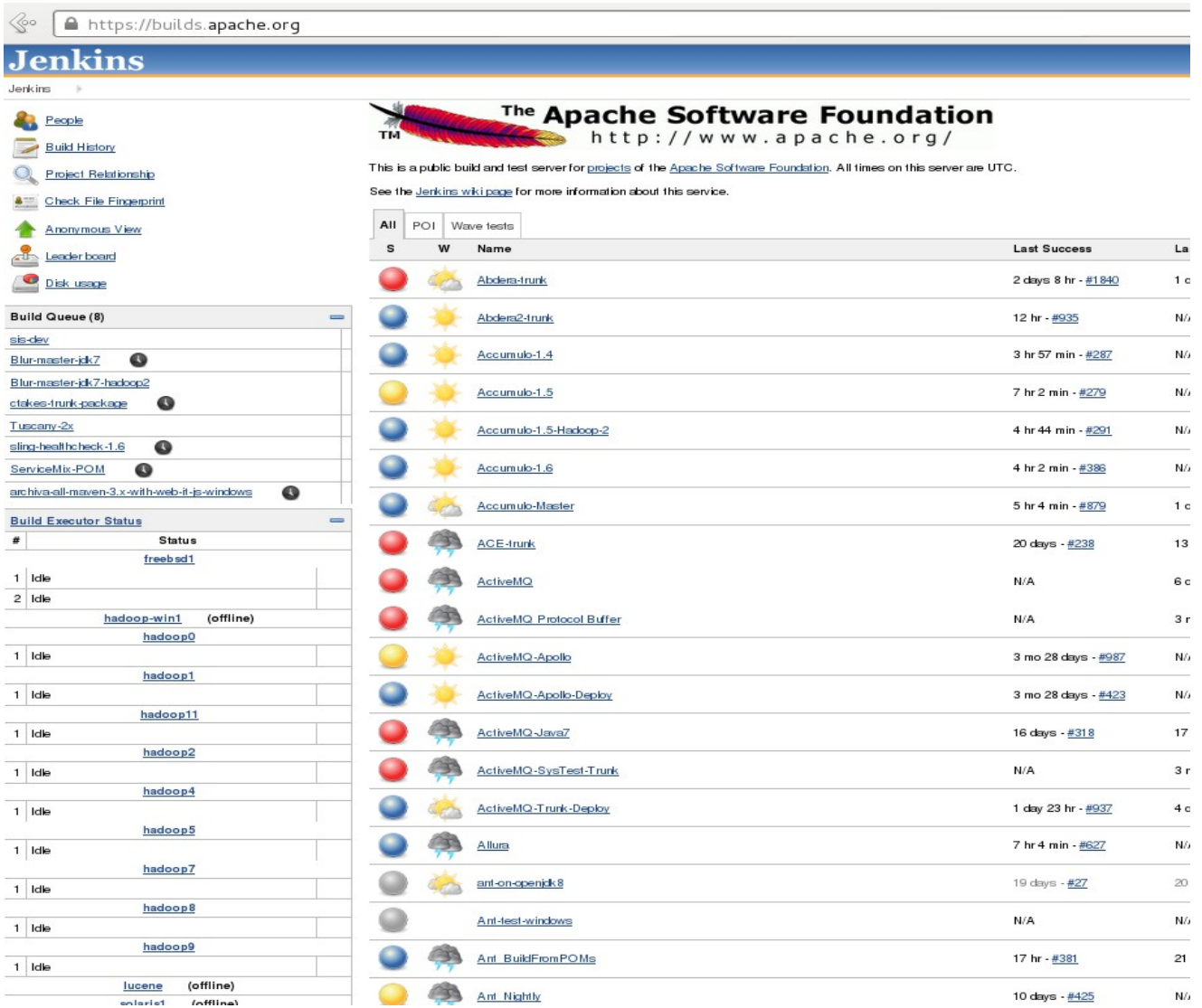
All | Buckminster | EMF | Eclipse and Equinox | EclipseLink | Hudson | Jetty-RT | Modeling | Mylyn | Repository Aggregation

S	W	Job ↓	Last Success	Last Failure
		aether-ant-nightly	19 hr (#34)	22 days (#31)
		aether-core-nightly	11 days (#323)	22 days (#322)
		amalgam-1.2.0	1 day 14 hr (#300)	22 days (#299)
		amp-integration	2 yr 11 mo (#24)	2 yr 10 mo (#59)
		amp-nightly	1 yr 11 mo (#1174)	1 yr 2 mo (#1188)
		babel-tools-indigo-nightly	11 mo (#56)	1 mo 4 days (#63)
		babel-tools-indigo-release	1 yr 2 mo (#3)	1 yr 2 mo (#2)
		bpel-helios	8 mo 7 days (#222)	1 mo 4 days (#232)
		bpel-juno	6 mo 8 days (#103)	1 mo 4 days (#115)
		bpmn2-nightly	8 mo 11 days (#15)	N/A
		buckminster-egf-helios	10 mo (#217)	N/A
		buckminster-egf-indigo	10 mo (#138)	N/A
		buckminster-egf-juno	12 mo (#67)	N/A
		buckminster-egf-kepler	N/A	N/A
		buckminster-egf-luna	N/A	N/A
		buckminster-egf-trunk-nightly	7 mo 13 days (#88)	N/A
		buckminster-emf-diffmerge-helios	11 mo (#30)	N/A
		buckminster-emf-diffmerge-master	6 days 5 hr (#50)	N/A
		buckminster-head	1 mo 4 days (#230)	N/A
		buckminster-maintenance	1 mo 20 days (#118)	N/A
		buckminster-sphinx-0.7-helios	1 yr 11 mo (#289)	N/A

~25 visible
out of ~300

Another Meteo ?

https://builds.apache.org/



~25 visible
out of >= 1000

Victory Celebration Pause



- [x] Test
- [x] Develop
- [x] Refactor

(Agile Pulse)

- [x] GIT - Team work
- [x] Jenkins Automated Build

Questions So Far ?

Full Victory ??



- [x] Test
- [x] Develop
- [x] Refactor

(Agile Pulse)

- [x] GIT - Team work
- [x] Jenkins Automated Build

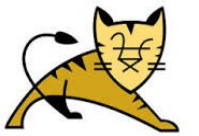
- [ ] Integration Tests / Real Apps
- [ ] Deliver to Envs : INT,UAT,TNH,PROD
- [ ] DevOps / Monitoring



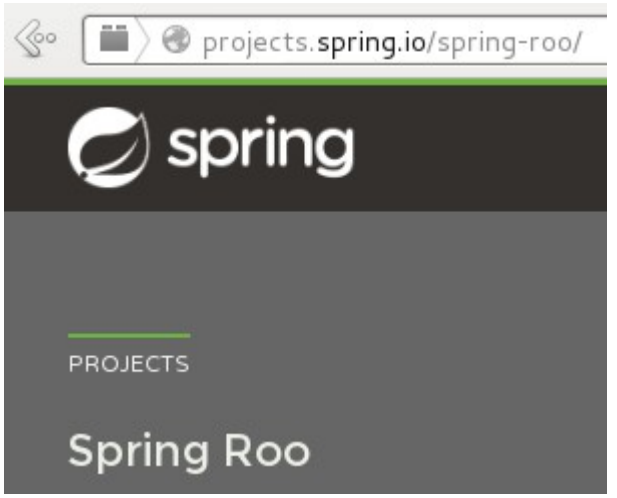

Creating a Real Java Web App

- Objective : run a WebApp != HelloWorld
  - on tomcat/jetty
  - Using Standard Modern techno
    Spring + JPA + Eclipselink (on server-side)

    Whatever on client-side (jspx, GWT, json+Angular)
  - Test .. Test … Test
- Constraint: <= 3 minutes for Demo time!
- Several alternatives … as “Wizards”




A WebApp in LESS Than 3 mn

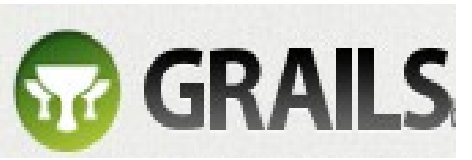




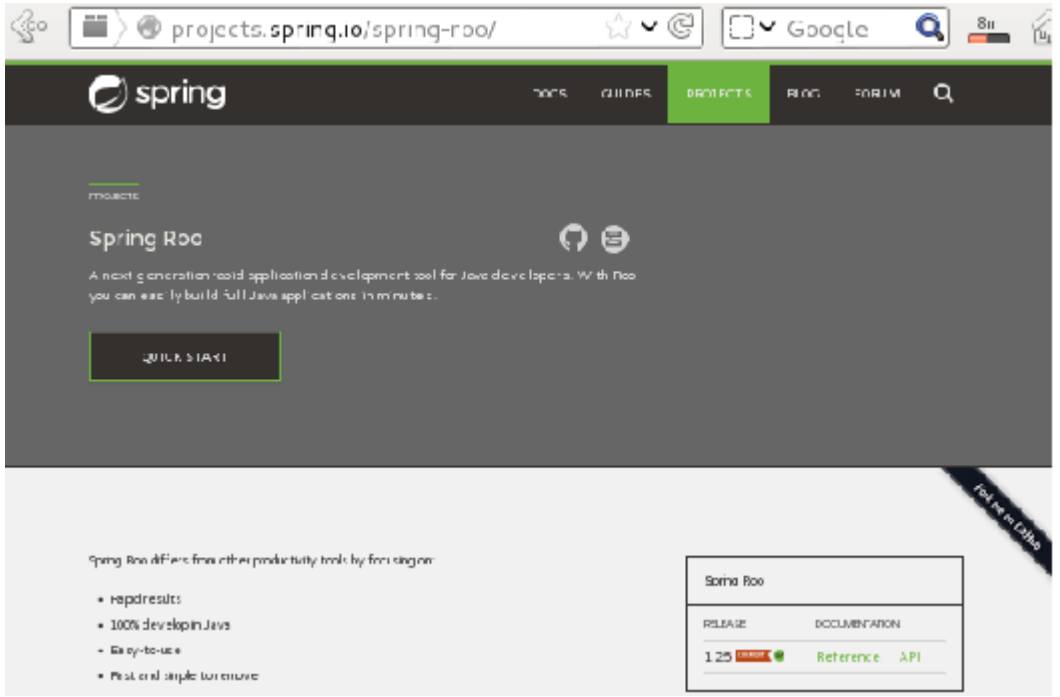
http://start.spring.io/







ROO - Get Hands Dirty





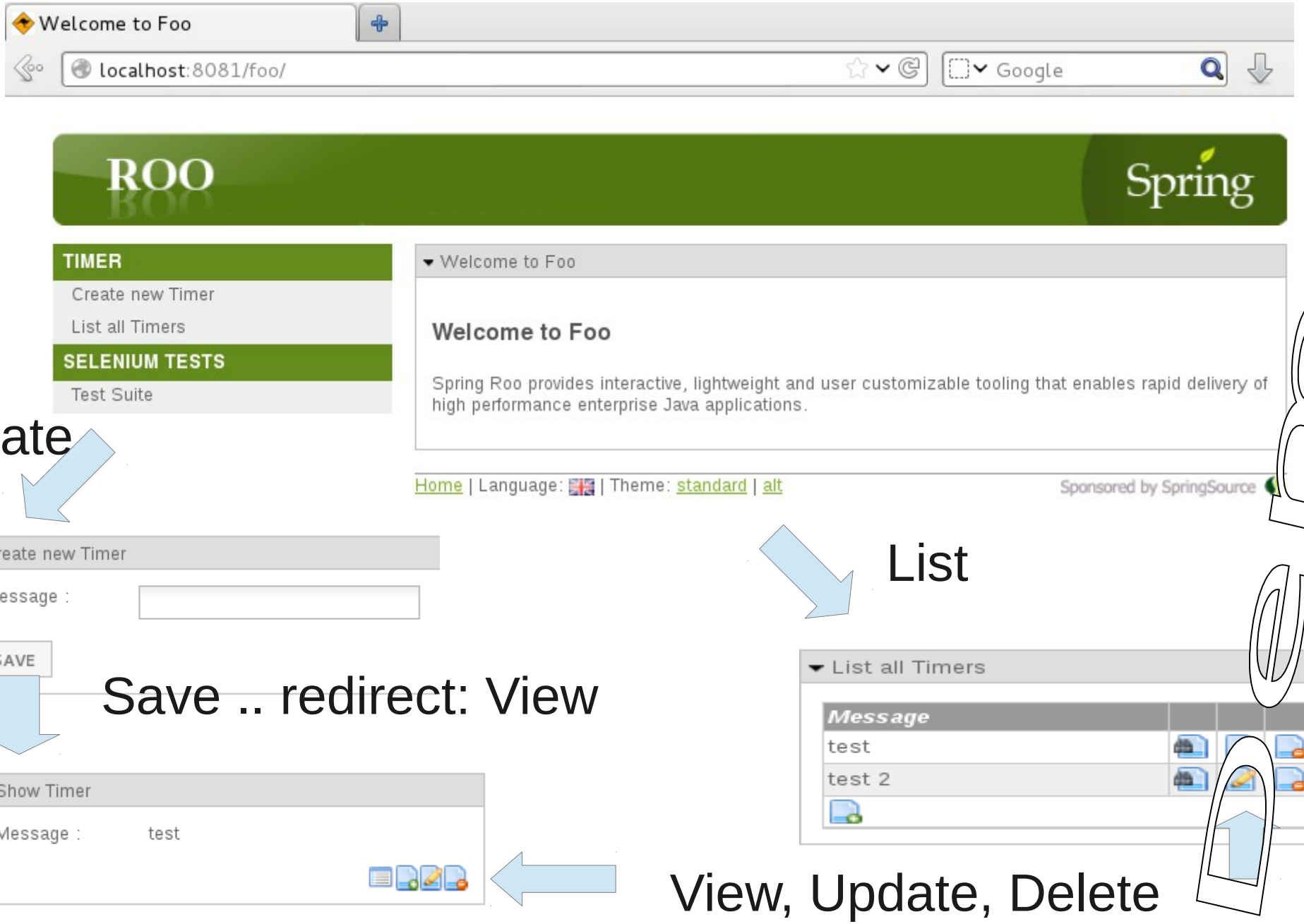
```
Welcome to Spring Roo.
roo>
```

```
mkdir hello
cd hello
roo
roo> hint
roo> project --topLevelPackage com.foo
roo> jpa setup --provider HIBERNATE --database HYP
roo> entity jpa --class ~.Timer --testAutomaticall
roo> field string --fieldName message --notNull
roo> hint web mvc
roo> web mvc setup
roo> web mvc all --package ~.web
roo> selenium test --controller ~.web.TimerController
roo> web gwt setup
roo> web gwt all --proxyPackage ~.client.proxy --requestPackage ~.client.reques
t
roo> perform tests
roo> quit
```

ZOOM

hint
project --topLevelPackage com.foo
jpa setup --provider HIBERNATE --database HYPERSONIC_IN_MEMORY
entity jpa --class ~.Timer --testAutomatically
field string --fieldName message --notNull
web mvc setup
web mvc all --package ~.web
selenium test –controller ~.web.TimerController

Launching Roo WebApp






Forgot TDD Test Promises ?

- [x] TDD in Hello World
  FAST dev env
  maven build
- [x] Real WebApp
- [x] launched + test .. MANUALLY only !!
- [ ] What's next ?
- [ ] Do same with real JEE project... Tomcat,Spring
  tackle complexity

Small < Medium < Big Tests
Integration Tests





Small Test : Real Unit Test
… using in memory (Mockito)



Medium Test : Integration Test with **White Box**
… (no real client / server, ex: Spring Junit)



Big Test : Integration Test with **Black Box**
… using Selenium + REAL server



Integration Tests

HTTP Client Integration Test (Medium test)






```xml
<dependency>
    <groupId>org.jsoup</groupId>
    <artifactId>jsoup</artifactId>
    <version>1.7.3</version>
    <scope>test</scope>
</dependency>
```

```java
public class TimerHttpIT {

    @Test
    public void testHttpGetIndex() throws Exception {
        // Perform
        Document doc = Jsoup.connect("http://localhost:8080/foo").get();
        Element titleElt = doc.select("head > title").get(0);
        String titleText = titleElt.text();

        // Post-check
        Assert.assertEquals("Welcome to Foo", titleText);
    }
}
```

Demo

Launching *IT.java Integration Test from Maven





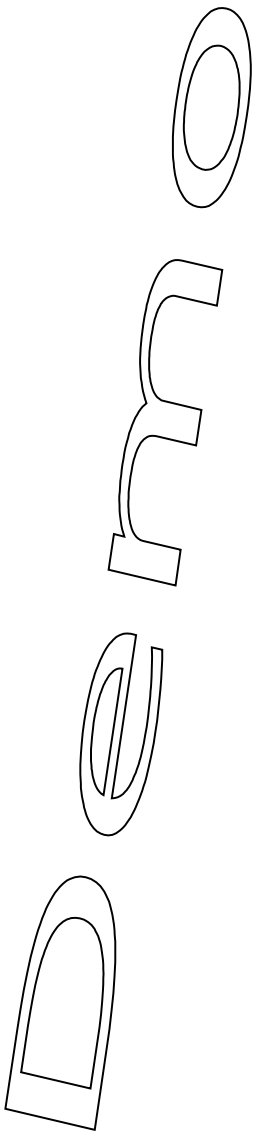
Integration-test
*IT.java

```
<plugin>
    <groupId>org.apache.maven.plugins</groupId>
    <artifactId>maven-failsafe-plugin</artifactId>
    <executions>
        <execution>
            <phase>integration-test</phase>
            <goals>
                <goal>integration-test</goal>
            </goals>
        </execution>
    </executions>
</plugin>

<plugin>
    <groupId>org.eclipse.jetty</groupId>
    <artifactId>jetty-maven-plugin</artifactId>
    <configuration>
        <webAppConfig>
            <contextPath>/foo</contextPath>
        </webAppConfig>
        <stopPort>10000</stopPort>
        <stopKey>STOP</stopKey>
    </configuration>
    <executions>
        <execution>
            <id>start-jetty</id>
            <phase>pre-integration-test</phase>
            <goals>
                <goal>run</goal>
            </goals>
            <configuration>
                <scanIntervalSeconds>0</scanIntervalSeconds>
                <daemon>true</daemon>
            </configuration>
        </execution>
        <execution>
            <id>stop-jetty</id>
            <phase>post-integration-test</phase>
            <goals>
                <goal>stop</goal>
            </goals>
        </execution>
    </executions>
```

Pre-integration-test
= start

Post-integration-test
= stop



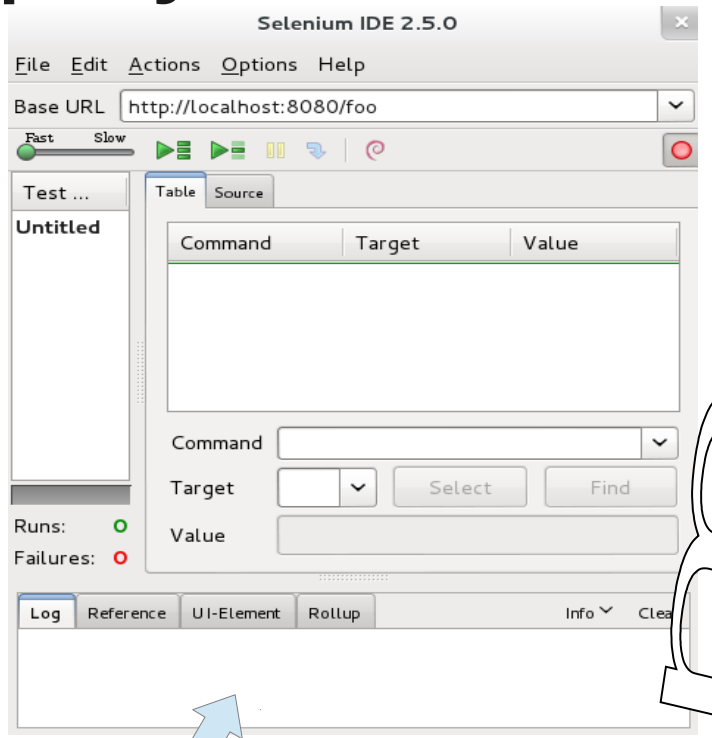
Big Test – Real Browser + Selenium Record-Replay

Clue 1 … From ROO script

```
roo> selenium test --controller ~.web.TimerController
Updated SRC_MAIN_WEBAPP/selenium/test-timer.xhtml
Updated SRC_MAIN_WEBAPP/selenium/test-suite.xhtml
```

Clue 2 … From WebApp







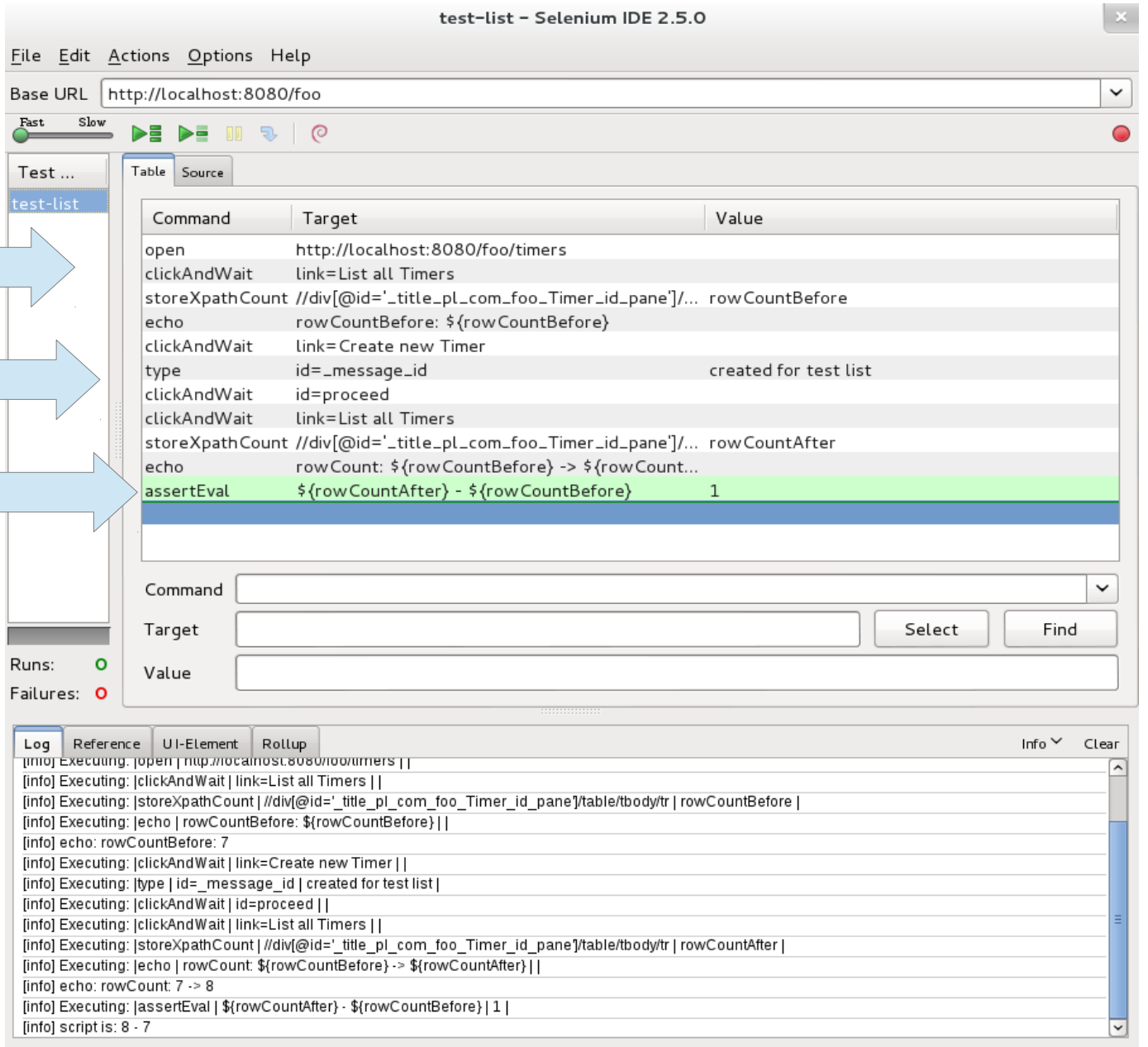
Selenium test for TimerController		
open	http://localhost:8081/foo/timers?form&lang=en_US	
type	_message_id	someMessage1
clickAndWait	//input[@id = 'proceed']	
verifyText	_s_com_foo_Timer_message_message_id	someMessage1

Selenium Test

list count before

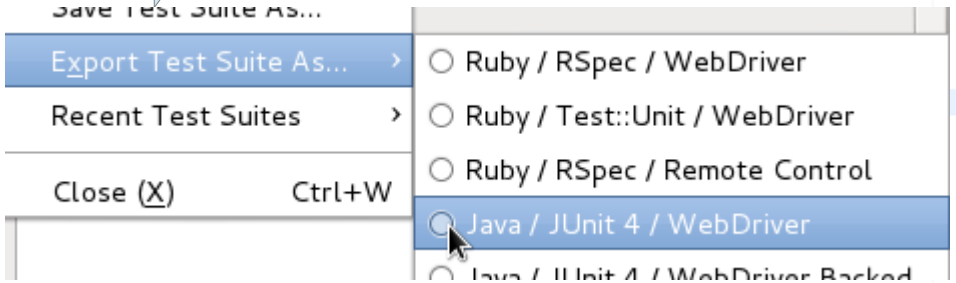
Save new

list count after Assert +1



Selenium Test Exported in Java

export



Add dependency

```
<dependency>
    <groupId>org.seleniumhq.selenium</groupId>
    <artifactId>selenium-firefox-driver</artifactId>
    <version>${selenium.version}</version>
    <scope>test</scope>
</dependency>
```

Refactor test

TimerSeleniumIT.java

```
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.firefox.FirefoxDriver;

public class TimerSeleniumIT {
    private WebDriver driver;
    private String baseUrl = "http://localhost:8080/foo";
    private StringBuffer verificationErrors = new StringBuffer();

    @Before
    public void setUp() throws Exception {
        driver = new FirefoxDriver();
        driver.manage().timeouts().implicitlyWait(30, TimeUnit.SECONDS);
        driver.get("http://localhost:8080/foo");
        System.out.println("Done init FirefoxDriver");
    }

    @Test
    public void testTimerSeleniumIT() throws Exception {
        System.out.println("testTimerSeleniumIT ...");
        doSaveNewTimer("created for test list 1");

        System.out.println("click List All timers...");
        driver.get(baseUrl);
        driver.findElement(By.linkText("List all Timers")).click();
        int rowCountBefore = driver.findElements(By.xpath("//div[@id
        System.out.println("rowCountBefore: " + rowCountBefore);

        doSaveNewTimer("created for test list 2");
        doSaveNewTimer("created for test list 3");

        driver.get(baseUrl);
        driver.findElement(By.linkText("List all Timers")).click();
        int rowCountAfter = driver.findElements(By.xpath("//div[@id

        System.out.println("rowCount: " + rowCountAfter + " (was
        Assert.assertEquals(rowCountBefore + 2, rowCountAfter);
    }

    private void doSaveNewTimer(String timerName) {
        System.out.println("click Create New Timer...");
        driver.get(baseUrl);
        driver.findElement(By.linkText("Create new Timer")).click();
        driver.findElement(By.id("_message_id")).clear();
        driver.findElement(By.id("_message_id")).sendKeys(timerName);
        driver.findElement(By.id("proceed")).click();
    }
```

Results Automated Test Start + Test + Stop






Launch

```
$
$ mvn install  -P integration -DskipTests=false
[INFO] Scanning for projects...
```

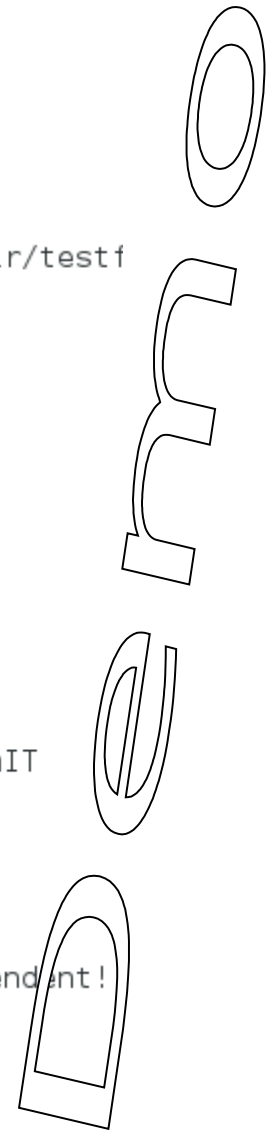
```
[INFO] Started Jetty Server
[INFO]
[INFO] --- maven-failsafe-plugin:2.17:integration-test (default) @ foo ---
[INFO] Failsafe report directory: /mnt/a_1tera2/homeData/arnaud/perso/devPerso/continous-int/demo-dir/testf

-------------------------------------------------------
 T E S T S
-------------------------------------------------------
Running com.foo.TimerSeleniumIT
Done init FirefoxDriver
testTimerSeleniumIT ...
click Create New Timer...
click List All timers...
rowCountBefore: 2
click Create New Timer...
click Create New Timer...
rowCount: 4 (was 2)
Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 5.447 sec - in com.foo.TimerSeleniumIT

Results :

Tests run: 1, Failures: 0, Errors: 0, Skipped: 0

[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO]
[INFO] --- jetty-maven-plugin:9.2.0.M0:stop (stop-jetty) @ foo ---
[INFO]
```



Real Java WebApp
Integration Tests
Victory Celebration





Integration Tests / Real Apps

Questions so far ?

Real Java WebApp
Integration Tests
Victory Celebration

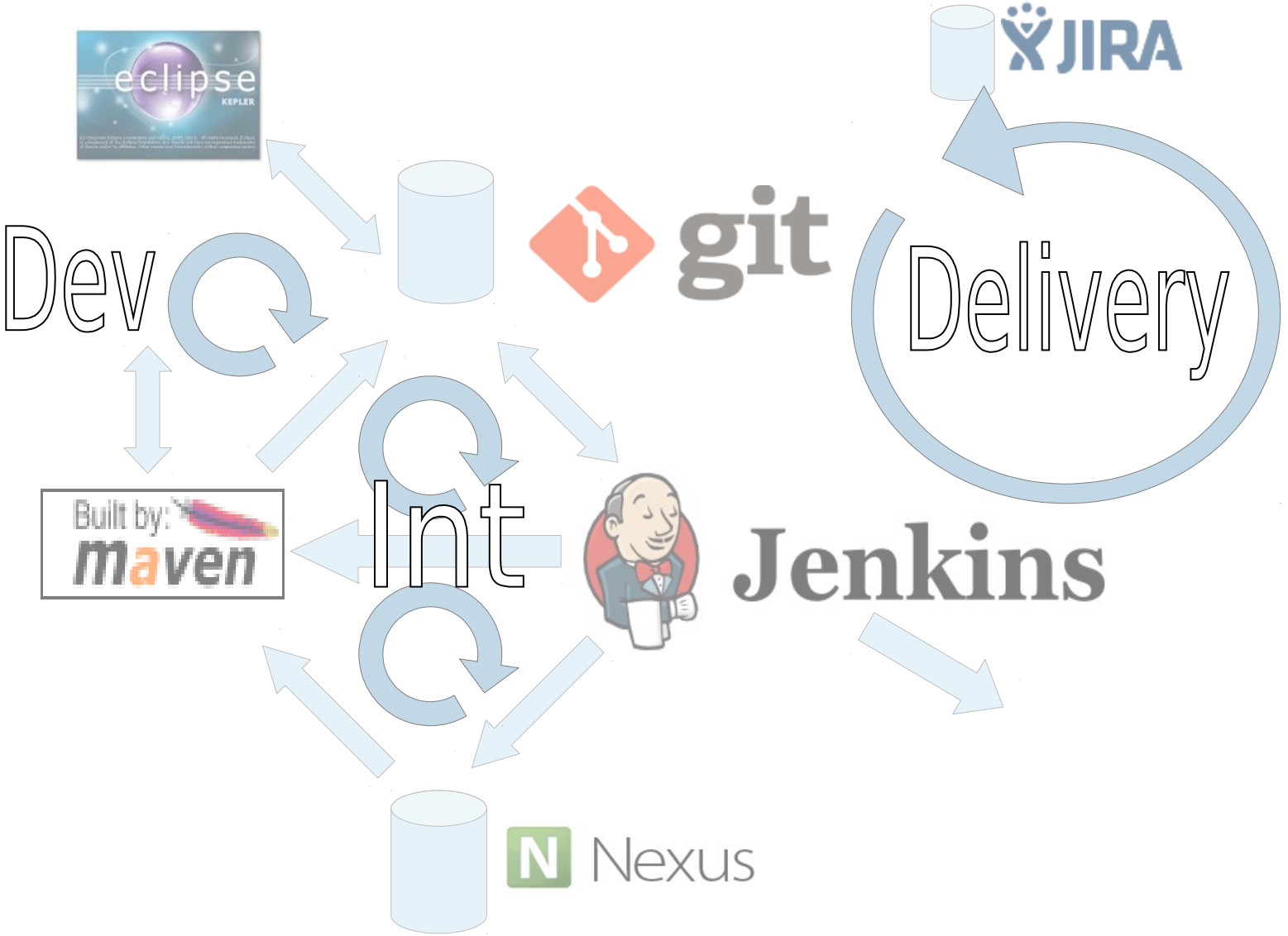


- [x] Integration Tests / Real Apps

next ?

- [ ] Deliver to Envs : INT,UAT,TNH,PROD
- [ ] DevOps / Monitoring

Workflow Reminder …
Dev → Integration → Delivery



XL Deploy (XebiaLabs (c))